

BECM 3115

CLIMATE & ARCHITECTURAL DESIGN

COMPLETE QUESTION BANK SOLUTION

2016 - 2024 Regular Final Papers

Organization

Section A and Section B are regrouped according to the current set-distribution system used in your preparation: 4 sets per section. Historical papers sometimes placed the same topic under the opposite section; the original year/question reference is preserved beside every solution.

Completeness rule

Repeated questions are consolidated into master answers so the same explanation is not duplicated unnecessarily. A paper-by-paper coverage checklist at the end maps every question/sub-question to a solution ID. Numerical variants are solved separately.

Primary basis: uploaded class notes + lecture slides + regular final question bank + uploaded reference books for clarification.

Made by 2223033

How to Use This Solution Book

3-out-of-4 strategy

Prepare at least 3 sets strongly from each section. The order inside every set is most-repeated first; S-priority answers are the safest high-return blocks.

Answer style used

- Question variants and years
- Core definition
- Step-by-step explanation
- Tables / sketches / formulas
- Exam-ready conclusion
- Bangla memory note where a concept is difficult

Source discipline and correction policy

- Course slides/class notes are treated as the primary exam source.
- Where a question asks something the uploaded slides only name but do not fully derive (for example the full solar-shading numerical chain, U-value/time-lag equations, bioclimatic chart detail or ecological-design expansion), that part is explicitly marked "Supplementary standard method".
- Several repeated fire-shelter/laboratory question scans reorder the ventilation-table headings. The solution uses the consistent lecture-slide table: at 3 m³/person = Minimum 11.3 L/s, Smoking 22.6 L/s, Non-smoking 17.0 L/s.
- For ventilation heat-loss numericals, the paper gives specific heat but not air density. Results are therefore shown symbolically and then numerically with the stated standard assumption $\rho = 1.20 \text{ kg/m}^3$.

Set Distribution

Section	Set	Core cluster	Priority
A	A1	Thermal Comfort, Heat Exchange & Comfort Indices	S
A	A2	Ventilation, Air-space, Infiltration & Numericals	S
A	A3	Microclimate, Envelope, Airflow & Shading	S
A	A4	Climate-Context + Climate-friendly Construction	A/S
B	B1	Global Climate, Earth-Sun, Thermal Balance & Winds	S
B	B2	Tropical Climate Classification & Design Response	S
B	B3	Regional, Site & Urban Climate	S
B	B4	Elements/Measurement/DRI + Basic Thermal Response	A/S

SECTION A - COMPLETE SOLUTIONS | 4-set distribution

Preparation priority

Safest order: A1 → A2 → A3 → A4. A2 gives guaranteed numerical practice; A3 gives sketch-heavy design marks.

SET A1 - Thermal Comfort, Heat Exchange & Comfort Indices

Set logic

This is the most repeated theory family. Always draw a human heat-exchange sketch and write the heat-balance logic before expanding into special climatic conditions.

A1.1 Human Body Heat Exchange: conduction, convection, radiation & evaporation

Priority: S - very high | Asked: 2016, 2018, 2020, 2024/related | Marks: 15-20

Year	Q.	Marks	Question focus
2016	5(b)	20	Explain the human body heat exchange process.
2018	4(b)	15	Human body heat exchange with necessary sketches.
2020	1(a)	15	Heat exchange with environment through conduction, convection, radiation and evaporation.
2018	7(a)	20	Thermal comfort: production and release of heat to environment (basic overlap).

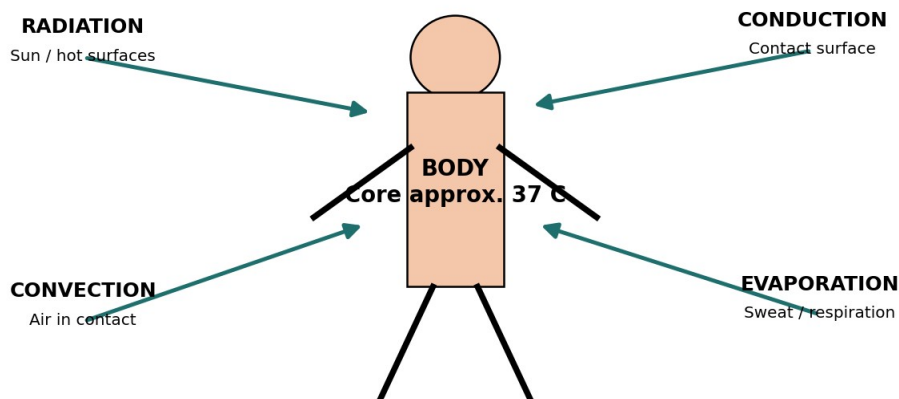
Exam answer structure

Definition Metabolism / heat gain e 4 xchange paths Heat -balance link S ketch Design implication

The human body is a heat-producing system. Food is converted into energy by metabolism; only part is used as useful work and the remaining heat must be transferred to the environment so that deep-body temperature remains close to its normal level.

Process	How heat is transferred	Example / controlling factor
Conduction	Direct molecular heat transfer through physical contact.	Feet on a cool floor; clothing/seat contact. Usually a smaller whole-body path.
Convection	Heat exchange between skin/clothing and adjacent air. Warm air rises and is replaced by cooler air.	Higher air speed and larger skin-air temperature difference increase convective exchange.
Radiation	Long-wave exchange between the body and surrounding surfaces without direct contact.	Hot roof/wall or sun can add radiant load; cool surrounding surfaces increase radiant loss.
Evaporation	Latent heat is removed when sweat/moisture evaporates; also through respiration/perspiration.	Most important when air is hot; effectiveness falls as humidity approaches saturation.

Human heat exchange: heat can be gained or lost depending on temperature gradients



Human-body heat exchange -redraw this in the exam.

$$M - W = C + R + K + E + S$$

At steady state $S \approx 0$. M = metabolism; W = useful work; C = convection; R = radiation; K = conduction; E = evaporation; S = body heat storage.

বাংলায় বুঝুন

শরীর যত net heat তৈরি করে, তত heat পরিবেশে ছাড়তে না পারলে শরীরে heat জমে (S positive) এবং শরীর গরম হয়। বেশি heat হারালে S negative হয়ে ঠান্ডা লাগে।

Conclusion

A comfortable building does not “cool the body” by one mechanism only; it manages air temperature, surface temperature, humidity and air movement so that the four heat-exchange paths remain physiologically manageable.

A1.2 Thermal / Climatic Comfort: factors, ideas and significance

Priority: S - very high | Asked: 2016, 2018, 2023, 2024 | Marks: 5-20

Year	Q.	Marks	Question focus
2016	6(a)	15	Factors of Climate Comfort with examples.
2016	6(b)	5	Thermal comfort ideas.
2018	7(a)	20	What is thermal comfort? How does body produce/release heat?
2023	4(c)	10	Significance of thermal comfort for human body.
2024	3(c)	12	What is thermal comfort? How does body maintain thermal balance?

Thermal comfort is the condition in which a person feels thermally satisfied and the body can maintain heat balance without excessive physiological effort such as heavy sweating or shivering.

Objective climatic variable	Effect on comfort
Air temperature	Controls sensible heat exchange, especially convection.
Mean radiant / surrounding surface temperature	Controls radiant heat gain/loss from roof, walls, sun and other surfaces.
Humidity	Controls the air's capacity to accept evaporated moisture; high RH reduces sweat evaporation.
Air movement	Changes convective and evaporative heat exchange and therefore perceived comfort.

Subjective / individual variables

- Clothing
- Acclimatization
- Age and sex
- Body shape and subcutaneous fat
- State of health
- Food and drink
- Skin colour / individual variation

Memory

Objective = T-R-H-A: Temperature, Radiation, Humidity, Air movement. Subjective = person-specific variables.

Conclusion

Thermal comfort matters because sustained thermal imbalance reduces health, concentration and performance; climate-responsive design therefore aims to reduce physiological strain before mechanical conditioning is added.

A1.3 Human Thermal Balance Equation & physiological regulation

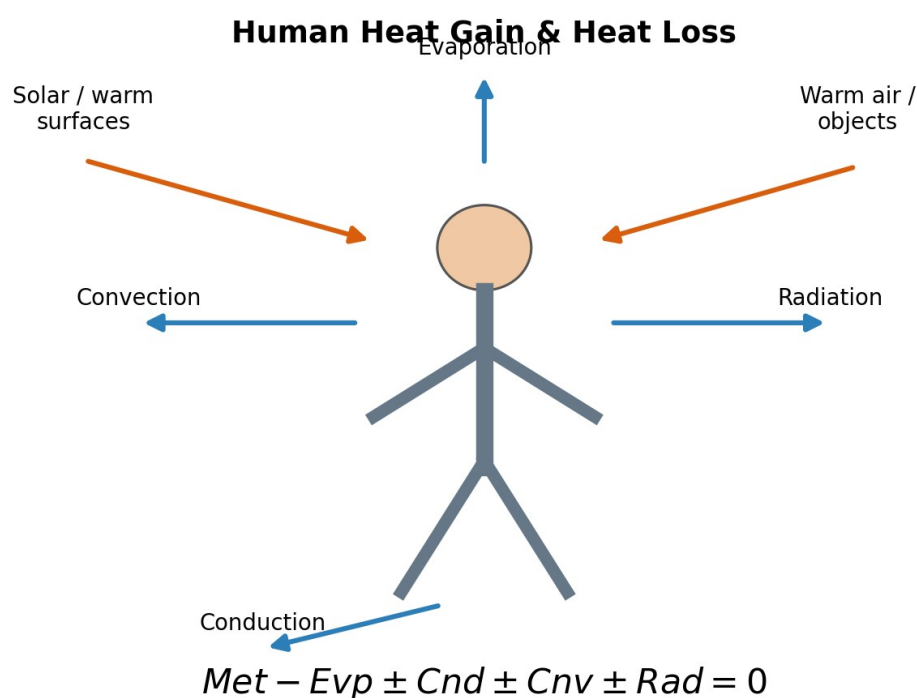
Priority: S - high | Asked: 2016, 2021, 2023, 2024 | Marks: 5-13

Year	Q.	Marks	Question focus
2016	7(b)	5	What is the thermal balance equation?
2021	4(a)	13	Write human thermal balance equation; what happens if unbalanced?
2023	4(d)	12	Thermal balance equation and regulatory mechanisms when unbalanced.
2024	3(c)	12	How does human body maintain thermal balance?

$$M - W = C + R + K + E + S$$

For thermal equilibrium, heat produced and gained must equal heat lost. At steady state, $S \approx 0$. A positive storage term means body heat is accumulating; a negative storage term means the body is losing stored heat.

Imbalance	Body response	Purpose
Too warm / $S > 0$	Vasodilation → more blood to skin; then sweating if needed.	Increase sensible and evaporative heat loss.
Too cold / $S < 0$	Vasoconstriction → less blood to skin; shivering if needed.	Reduce heat loss and increase metabolic heat production.



Human thermal balance - central equation with regulatory response.

Exam line

If vasomotor regulation is insufficient in heat, sweating begins; in cold, strong shivering can greatly increase metabolic heat production.

A1.4 Comfort Indices: ET, CET, EW, OT + effective-temperature limitations

Priority: S - very high | Asked: 2016, 2017, 2018, 2021, 2022, 2023 | Marks: 5-15

Source status

The uploaded question bank names this topic, but the uploaded slides/notes do not contain the full derivation or numerical formula chain. The marked part below uses standard building-physics / solar-geometry knowledge as a supplementary method.

Year	Q.	Marks	Question focus
2016	6(c)	15	ET, CET and bioclimatic chart.
2017	7(a-v)	part of 25	Corrected Effective Temperature.
2018	7(c)	10	Effective temperature and its limitations.
2021	6(c)	10	Differentiate ET, CET and equivalent warmth.
2022	6(c)	10	Progressive development of ET, CET, EW and OT.
2023	5(b)	12	Progressive development of ET, CET, EW and OT and integrated factors.

Index	Main variables / idea	Strength / limitation
ET - Effective Temperature	Integrates air temperature + humidity + air movement into an equivalent sensation scale.	Radiant environment is not directly included.
CET - Corrected Effective Temperature	ET corrected by including radiation effect.	More complete than ET; widely used in the course material.
EW - Equivalent Warmth	Uses air temperature, humidity, mean radiant temperature and skin/clothing surface response.	Can underestimate cooling effect of air movement in high humidity.
OT - Operative Temperature	Combines air temperature and mean radiant temperature.	Most useful where humidity and air movement effects are small/negligible.

Progressive development

1. ET: first combine temperature + humidity + air movement.
2. CET: add radiation to correct ET.
3. EW: correlate environmental variables with skin/clothing response.
4. OT: simplify to air temperature + mean radiant temperature when moisture and air movement are not dominant.

ET limitations

- Does not directly represent radiant asymmetry / mean radiant temperature.
- One index cannot capture all individual differences, clothing and acclimatization.
- Comfort response changes with activity, air speed and humidity range; therefore ET is an index, not a universal physiological law.

Bioclimatic chart - supplementary description

The uploaded Section A material asks the bioclimatic chart but does not provide its full construction. Exam-safe interpretation: it is a climate-comfort chart that plots climatic variables (commonly temperature with humidity) and shows a comfort zone plus regions where strategies such as air movement, shading or evaporative response may extend comfort. Draw only the version taught in class if your teacher supplied a specific chart.

A1.5 Heat loss / heat gain in different climatic situations

Priority: S - highest repeat | Asked: 2018, 2020, 2021, 2022, 2023 | Marks: 12-20

Year	Q.	Marks	Question focus
2018	4(a)	20	Processes of heat loss in various climatic situations.
2020	4(b)	15	Heat loss in various climatic situations.
2021	5(b)	12	Heat loss in various climatic situations.
2022	5(b)	12	Heat loss in various climatic situations.
2023	6(c)	14	Heat loss or gain in various climatic conditions.

Situation	What happens	Dominant design lesson
Calm warm air + moderate RH	Body can dissipate surplus heat by radiation, convection and evaporation. Lecture example $\approx$ 45% radiation, 30% convection, 25% evaporation.	Comfort possible with modest air movement and moderate surface temperature.
Hot air + considerable radiation	As air temperature approaches skin temperature, convection falls; hot surroundings can add radiant heat.	Shade hot surfaces and keep air dry enough for evaporation.
Hot air + radiation + air movement	If $RH < 100\%$, moving air replaces the moist boundary layer around skin and accelerates evaporation.	Air movement gives strong relief even when sensible cooling is weak.
Air warmer than skin + saturated + still	Convection may add heat and evaporation becomes ineffective.	Dangerous heat-storage condition; strong ventilation / dehumidification / radiant control needed.

Correction note

Air movement does not "increase RH"; it increases evaporation by removing the saturated boundary layer when the surrounding air is not saturated.

A1.6 Designer responsibilities for thermal comfort

Priority: S - high | Asked: 2017, 2021, 2022, 2024 | Marks: 10-20

Year	Q.	Marks	Question focus
2017	6(c)	20	Responsibilities of built-form designers; complexities in climate design.
2021	5(c)	10	Designer responsibilities in ensuring thermal comfort.
2022	5(a)	10	Designer responsibilities + design tools for moderating microclimate.
2024	6(b)	10	Designer responsibilities and domains of essential knowledge.

External responsibilities	Internal responsibilities
Minimum interference to useful normal wind flow.	Provide adequate oxygen-rich air.
Control adverse solar impact / harmful glare.	Maintain conducive indoor temperature.
Manage precipitation and driving-rain exposure.	Maintain suitable humidity, air quality and airflow.

How responsibilities become design decisions

- Orient form and openings to useful wind and solar geometry.
- Use envelope thickness, insulation, reflectance and thermal capacity strategically.
- Use shading before mechanical cooling where possible.
- Provide cross/stack ventilation and adequate air-change paths.
- Use active systems only when passive control cannot maintain acceptable conditions.

A1.7 Passive vs Active Thermal Control

Priority: A - repeated | Asked: 2018, 2020, 2022/related | Marks: 10-15

Year	Q.	Marks	Question focus
2018	8(a)	10	Distinguish passive and active thermal control; which do you prefer?
2020	4(a)	15	Differentiate active and passive thermal control; preference for best indoor condition.
2022	5(a)	part	Designer responsibilities and microclimate tools - passive/active logic.

Passive control	Active control
Uses form, orientation, shading, thermal mass, insulation, vegetation and natural ventilation.	Uses mechanical/electrical systems: fans, HVAC, mechanical ventilation, heaters, dehumidifiers.
Lower operational energy; climate-responsive; may have seasonal limits.	High controllability; higher energy, cost and maintenance; depends on services.
Should be the first design layer.	Should supplement passive design when comfort cannot otherwise be maintained.

Conclusion

A BECM professional should prefer a passive-first, active-assisted strategy rather than an absolute passive-only or active-only position.

A1.8 Essential knowledge for Climate Design

Priority: A - support answer | Asked: 2017, 2024 | Marks: 10

Year	Q.	Marks	Question focus
2017	6(b)	10	Essential knowledge for Climate Design.
2024	6(b)	part of 10	Domains of essential knowledge a designer should have.

Knowledge domain	Why needed
Science of climate	Understand temperature, radiation, wind, humidity and rainfall.
Physiology	Understand body response to heat/cold.
Human comfort	Set acceptable environmental targets.
Building materials	Select thermal capacity, insulation, moisture and durability response.
Passive design technology	Use orientation, form, shading, ventilation and landscape.
Active design technology	Use mechanical conditioning when passive measures are insufficient.

Memory

Climate + Body + Comfort + Material + Passive + Active.

A1.9 Vasomotor Adjustment & Autonomic Nervous System

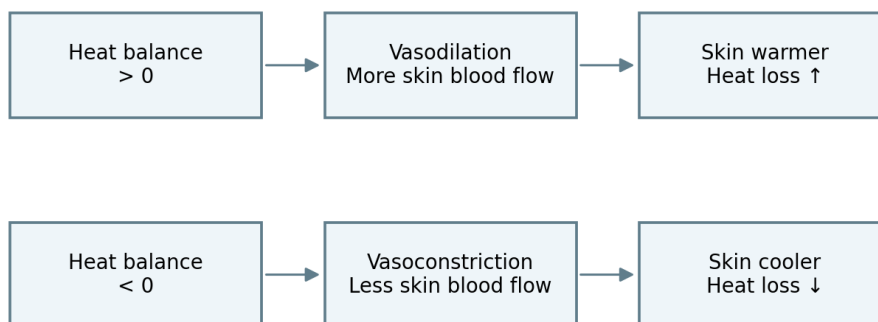
Priority: A/S - recent repeat | Asked: 2022, 2023, 2024 | Marks: 6-10

Year	Q.	Marks	Question focus
2022	6(d)	6	Vasomotor adjustment; autonomic nervous system dilate/constrict in ambient temperatures.
2023	5(a)	8	Vasomotor adjustment in different climates.
2024	6(a)	10	Vasomotor adjustment and design implications for comfortable indoor environments.

Vasomotor adjustment is the automatic change of blood-vessel diameter, especially near the skin, used by the autonomic nervous system to regulate heat transfer between the body core and the environment.

Condition	Response	Thermal effect
Warm / overheating	Vasodilation: skin blood vessels widen.	More warm blood reaches skin → skin temperature rises → greater convective/radiant loss; sweating may follow.
Cold / underheating	Vasoconstriction: skin blood vessels narrow.	Less warm blood reaches skin → skin temperature falls → heat loss slows; shivering may follow.

Thermoregulatory response (exam flowchart)



If vasomotor control is insufficient: overheating → sweating; underheating → shivering

Vasodilation vs vasoconstriction - redraw as a simple flow diagram.

Design implication

- Avoid prolonged indoor conditions that force continuous sweating or shivering.
- Use air speed to aid evaporation in warm conditions, but avoid excessive cold drafts.
- Reduce radiant heat from roofs/walls so vasodilation does not have to compensate for large radiant gains.
- Allow adaptive variation rather than a perfectly fixed condition all day.

SET A2 - Ventilation, Air-space, Infiltration & Numericals

Set logic

Memorize the course table and unit conversions. The numerical wording changes, but the workflow is repeated almost unchanged.

A2.1 Air Change, Ventilation Rate, Air-space & Infiltration - definitions

Priority: S - high | Asked: 2021, 2022, 2024/related | Marks: 5-8

Year	Q.	Marks	Question focus
2021	5(a)	5	Meaning of air change and how ventilation rate is expressed.
2022	8(d)	6	Short notes on air space, air change and infiltration rate.
2024	7(a)	part	Meaning of air space and infiltration rate.
2021	6(a)	part	Air space and infiltration rate.

Ventilation is the controlled replacement of stale indoor air with fresh, oxygen-rich air. Its purposes are to supply oxygen and remove carbon dioxide, unwanted heat, moisture, smell and pollutants.

$$\text{ACH} = (Q \times 3600) / V$$

Q = ventilation flow in m³/s; V = room volume in m³.

$$\text{Air-space per person} = \text{Net room volume} / \text{Number of users}$$

Term	Exam-ready definition
Air change	Replacement of the room air by fresh air; rate can be expressed as flow quantity or air changes per hour.
Air-space	The net volume of built space available per user for healthy air dilution.
Infiltration	Uncontrolled entry of outdoor air through cracks, gaps, doors, windows and other leakage paths; the lecture uses the term for fresh-air entry rate in these numericals.
Ventilation rate	Either m ³ /s or L/s, and/or air changes per hour (ACH).

Course ventilation table used for all calculations

Air-space / person (m ³)	Minimum L/s	Smoking L/s	Non-smoking L/s
3	11.3	22.6	17.0
6	7.1	14.2	10.7
9	5.2	10.4	7.8
12	4.0	8.0	6.0

Important paper-scan inconsistency

Some fire-shelter/laboratory tables in the scanned question papers print 22.6 under "Minimum" and 11.3 under "Non-smoking". The lecture slide consistently gives 11.3 minimum, 22.6 smoking, 17.0 non-smoking at 3 m³/person. This solution follows the lecture table and flags the discrepancy instead of silently accepting the scan typo.

A2.2 Room ventilation numericals: m³/s, L/s and ACH

Priority: S - repeated numerical | Asked: 2021, 2022, 2023 | Marks: 7-8

Year	Q.	Marks	Question focus
2021	5(d)	8	45×23×3 m room, 200 people, smoking prohibited.
2022	5(d)	8	45×23×3 m room, 150 people, smoking prohibited.
2023	6(d)	7	45×25×3 m room, 180 people, smoking prohibited; table supplied.

Data note for 2023

The 2023 paper clearly states a 45×25×3 m room. It differs from the 45×23×3 m room used in 2021 and 2022, so it is calculated separately rather than forced into the earlier dimensions.

Workflow: $V = L \times B \times H \rightarrow \text{air-space/person} = V/N \rightarrow \text{choose the non-smoking rate from the course table} \rightarrow Q_{\text{total}} = \text{rate} \times N \rightarrow \text{convert L/s to m}^3/\text{s} \rightarrow \text{ACH} = Q \times 3600 / V$.

2021 solution - 200 persons

$V = 45 \times 23 \times 3 = 3105 \text{ m}^3$. Air-space/person = $3105/200 = 15.525 \text{ m}^3$. The table ends at $12 \text{ m}^3/\text{person}$, so use the highest listed non-smoking rate = 6.0 L/s/person (same convention used in the uploaded Section A class note).

$$Q = 200 \times 6.0 = 1200 \text{ L/s} = 1.20 \text{ m}^3/\text{s}$$

$$\text{ACH} = 1.20 \times 3600 / 3105 = 1.39 \text{ h}^{-1}$$

2022 solution - 150 persons

Air-space/person = $3105/150 = 20.70 \text{ m}^3 \rightarrow \text{use } 12 \text{ m}^3 \text{ table bracket} \rightarrow 6.0 \text{ L/s/person}$.

$$Q = 150 \times 6.0 = 900 \text{ L/s} = 0.90 \text{ m}^3/\text{s}$$

$$\text{ACH} = 0.90 \times 3600 / 3105 = 1.04 \text{ h}^{-1}$$

2023 solution - 180 persons

$V = 45 \times 25 \times 3 = 3375 \text{ m}^3$. Air-space/person = $3375/180 = 18.75 \text{ m}^3 \rightarrow \text{use the highest listed } 12 \text{ m}^3 \text{ bracket} \rightarrow 6.0 \text{ L/s/person}$.

$$Q = 180 \times 6.0 = 1080 \text{ L/s} = 1.08 \text{ m}^3/\text{s}$$

$$\text{ACH} = 1.08 \times 3600 / 3375 = 1.15 \text{ h}^{-1}$$

Table-selection caution

The table has no row above $12 \text{ m}^3/\text{person}$ and does not state an interpolation rule. The uploaded class note explicitly uses the 12 m^3 row for larger air-space values. Follow any different convention demonstrated by your teacher if provided.

A2.3 Heat loss due to air change - all repeated variants

Priority: S - highest numerical | Asked: 2021, 2022, 2023, 2024 | Marks: 8-10

Source status

The uploaded question bank names this topic, but the uploaded slides/notes do not contain the full derivation or numerical formula chain. The marked part below uses standard building-physics / solar-geometry knowledge as a supplementary method.

Year	Q.	Marks	Question focus
2021	6(a)	10	V=225 m ³ , Ti=25°C, To=15°C, 4 ACH.
2022	6(a)	9	V=225 m ³ , Ti=23°C, To=18°C, 5 ACH.
2023	5(d)	10	V=250 m ³ , Ti=25°C, To=15°C, 3 ACH.
2024	7(a)	8	V=225 m ³ , Ti=25°C, To=15°C, 4 ACH.

$$Q_v = nV/3600$$

Ventilation volume flow, m³/s.

$$\text{Heat-loss rate} = \rho Q_v c_p (T_i - T_o)$$

Use $c_p = 1.005 \text{ kJ/kg}\cdot\text{K} = 1005 \text{ J/kg}\cdot\text{K}$. Paper omits ρ ; numerical results below assume $\rho = 1.20 \text{ kg/m}^3$.

Paper	$Q_v \text{ (m}^3/\text{s)}$	$\dot{m} \text{ (kg/s)}$	$\Delta T \text{ (K)}$	Heat loss
2021 / 2024	0.2500	0.3000	10	3.015 kW
2022	0.3125	0.3750	5	1.884 kW
2023	0.2083	0.2500	10	2.513 kW

If you do not want to assume density

Leave the result as $\rho \times Q_v \times c_p \times \Delta T$ and state that the paper did not supply air density. This is scientifically cleaner than hiding an assumption.

A2.4 Fire-shelter safety numericals - complete repeated solution

Priority: S - repeated numerical | Asked: 2021, 2022, 2023, 2024 | Marks: 8-10

Year	Q.	Marks	Question focus
2021	7(a)	10	8×7×3 m shelter, designed 500, actual 550, available fresh air 6 L/s.
2022	7(d)	10	9×10×3 m shelter, designed 400, actual 450, available 6 L/s.
2023	7(c)	10	9×10×3 m shelter, 450 occupants, available 6 L/s.
2024	6(d)	8	8×7×3 m shelter, 550 occupants, available 6 L/s.

Safety requires both adequate air-space per person and adequate fresh-air flow. The question does not state smoking; use the “Minimum” column for a basic life-safety check, and mention that a non-smoking comfort target would be more demanding.

2021 / 2024: 8×7×3 m, 550 persons

$$V = 168 \text{ m}^3; \text{air-space/person} = 168/550 = 0.305 \text{ m}^3/\text{person}$$

This is far below the table minimum of 3 m³/person. Available fresh air per person = 6/550 = 0.0109 L/s/person, while even the minimum 3 m³ row requires 11.3 L/s/person.

$$\text{Minimum required total fresh air} = 550 \times 11.3 = 6215 \text{ L/s}$$

$$\text{Volume required to provide 3 m}^3/\text{person} = 550 \times 3 = 1650 \text{ m}^3$$

2022 / 2023: 9×10×3 m, 450 persons

$$V = 270 \text{ m}^3; \text{air-space/person} = 270/450 = 0.600 \text{ m}^3/\text{person}$$

$$\text{Minimum required total fresh air} = 450 \times 11.3 = 5085 \text{ L/s}$$

$$\text{Volume required to provide 3 m}^3/\text{person} = 450 \times 3 = 1350 \text{ m}^3$$

Safety verdict

Both repeated shelter cases are severely under-provisioned. Improve by reducing occupant load, increasing shelter volume, providing multiple rooms/compartments, and installing a much larger reliable fresh-air system with emergency power and protected inlets/outlets.

A2.5 Absolute Humidity, Saturation Humidity, Relative Humidity + supersaturated air

Priority: A/S - repeated | Asked: 2017, 2020, 2021 | Marks: 5-12

Year	Q.	Marks	Question focus
2020	2(b)	12	AH=10 g/m ³ and SH=17 g/m ³ at 20°C; calculate RH.
2021	6(d)	5	Same AH/SH/RH numerical.
2017	7(a-iv)	part	Short note: Super Saturated Air.

Term	Meaning
Absolute Humidity (AH)	Actual water-vapour amount per unit mass or volume of air, using the course convention.
Saturation Humidity (SH)	Maximum/equilibrium water-vapour amount air can hold at the same temperature.
Relative Humidity (RH)	Ratio of actual moisture to saturation moisture at the same temperature, expressed in percent.
Supersaturated air	A transient state with vapour content above equilibrium saturation; unstable and condensation tends to occur.

$$RH = (AH / SH) \times 100\%$$

$$RH = (10 / 17) \times 100 = 58.82\% \approx 58.8\%$$

A2.6 Laboratory fresh-air suitability - all repeated variants

Priority: S - repeated numerical | Asked: 2021, 2022, 2023, 2024 | Marks: 5

Year	Q.	Marks	Question focus
2021	8(c)	5	5×4×2.1 m lab, 20 users, fresh air 5 L/s; normal + power failure.
2022	8(c)	5	6×5×2.5 m lab, 20 users, 5 L/s.
2023	8(b)	5	6×5×2.5 m lab, 20 users, 5 L/s.
2024	5(d)	5	5×4×2.1 m lab, 20 users, 5 L/s.

Case 1: 5×4×2.1 m

$$V = 42 \text{ m}^3; \text{air-space/person} = 42/20 = 2.10 \text{ m}^3/\text{person}$$

2.10 m³/person is below the minimum 3 m³/person. Available total fresh air is only 5 L/s = 0.25 L/s/person. At the 3 m³ table row, minimum fresh air would be 11.3 L/s/person (226 L/s total); a non-smoking target is 17 L/s/person (340 L/s total). Therefore the lab is not suitable even with the fan running; during power failure the forced supply becomes zero and the situation is worse.

Case 2: 6×5×2.5 m

$$V = 75 \text{ m}^3; \text{air-space/person} = 75/20 = 3.75 \text{ m}^3/\text{person}$$

Air-space exceeds 3 m³/person, but the fresh-air supply is still only 0.25 L/s/person. Using the conservative 3 m³ bracket, the required flow is at least 226 L/s total (340 L/s for non-smoking). Thus the lab is still unsuitable; power failure removes the forced supply entirely.

Conclusion

For both laboratory variants, ventilation capacity - not just room volume - is the decisive failure. Add a properly sized ventilation system, emergency power / passive backup openings and verified outdoor-air paths.

SET A3 - Designing Microclimate, Building Envelope, Airflow & Shading

Set logic

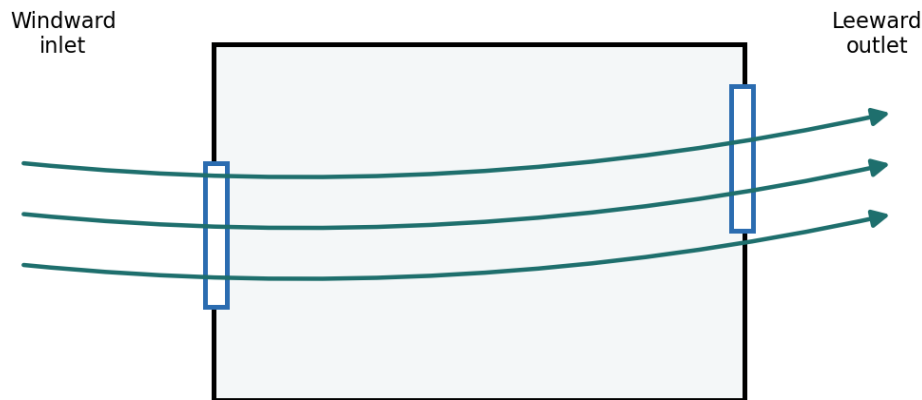
This set is sketch-heavy. A good answer should show at least one plan/section of airflow or solar control, not only text.

A3.1 Layout, spacing, opening position and cross ventilation

Priority: S - high | Asked: 2018, 2023, 2024/related | Marks: 8-20

Year	Q.	Marks	Question focus
2018	2(b)	20	Layout, spacing and opening position for air movement and ventilation.
2023	6(b)	8	Natural ventilation; spacing between blocks; methods of cross ventilation.
2021/22/23	7(b/a)	10	Microclimatic solution for hot-dry and warm-humid climates in terms of spacing.

Cross ventilation: openings on opposite / adjacent pressure zones



Good practice: keep airflow path unobstructed; outlet can be similar or slightly larger than inlet

Cross ventilation - inlet and outlet should create a clear path through the occupied zone.

Design variable	Warm-humid strategy	Hot-dry / composite strategy
Layout	Open/staggered forms to expose more façades to breeze.	Compact grouping to reduce hot exposed area; shaded courtyards can work.
Spacing	Broad spacing prevents wind shadow and lets breeze penetrate.	Closer spacing may create mutual shade and protected outdoor space.
Orientation	Present openings / building face to useful prevailing wind; slight angle can distribute flow.	Control hot dusty winds; use seasonal/selective openings.
Openings	Large, opposite or pressure-zone openings for cross flow; louvers direct air to occupied zone.	Smaller/controlled openings in hottest period; night ventilation where useful.

How cross ventilation operates

5. Wind creates positive pressure on windward side and negative/suction pressure on leeward side.
6. Place inlet and outlet on different pressure zones.
7. Keep an unobstructed internal flow path; avoid deep partitions that block the occupied zone.
8. Outlet size and height can be adjusted to improve extraction; high outlet also assists buoyancy/stack effect.

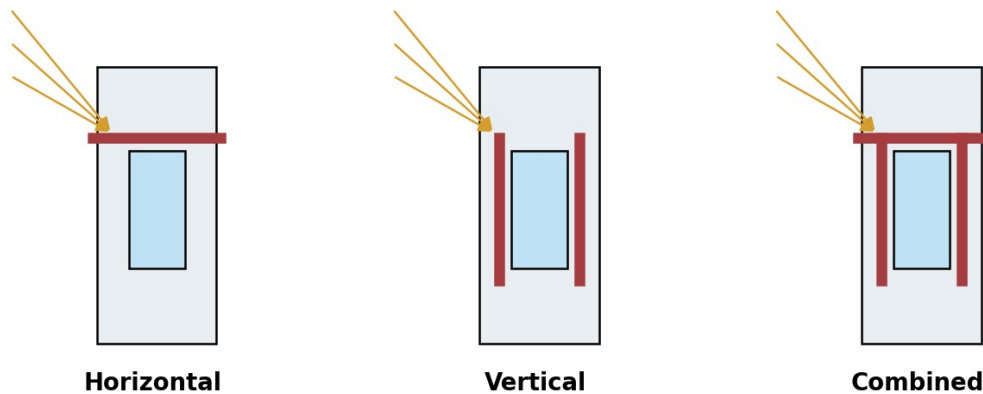
A3.2 Effective shading and types of shading devices

Priority: S - very high | Asked: 2016, 2021, 2022, 2024/related | Marks: 5-15

Year	Q.	Marks	Question focus
2016	7(c)	10	Effective shading with sketches.
2016	8(a)	15	Types of shading devices with sketches.
2021	8(a)	12	Ensure indoor climatic comfort by designing shading devices.
2022	8(a)	10	Ensure climatic comfort by shading devices.

Effective shading stops unwanted solar radiation before it becomes an indoor heat load while retaining useful daylight, view and seasonal sun where desirable. External shading is generally thermally more effective than internal curtains because it intercepts solar energy before it passes through glazing.

Basic shading-device types



Horizontal, vertical and combined shading devices.

Type	Best conceptual use	Examples
Horizontal	High-angle sun; often effective on south-oriented openings in the northern hemisphere.	Overhang, canopy, horizontal louvers.
Vertical	Low-angle east/west sun.	Vertical fins, side screens.
Combined	When both vertical and horizontal solar angles require control.	Egg-crate, fins + overhang.

Other climate-control functions

- Reduce direct and reflected solar radiation.
- Reduce glare.
- Protect wall/window from driving rain.
- Control privacy/visibility.
- Guide or partially screen wind where required.

A3.3 Cosine Law of Solar Incidence and building applications

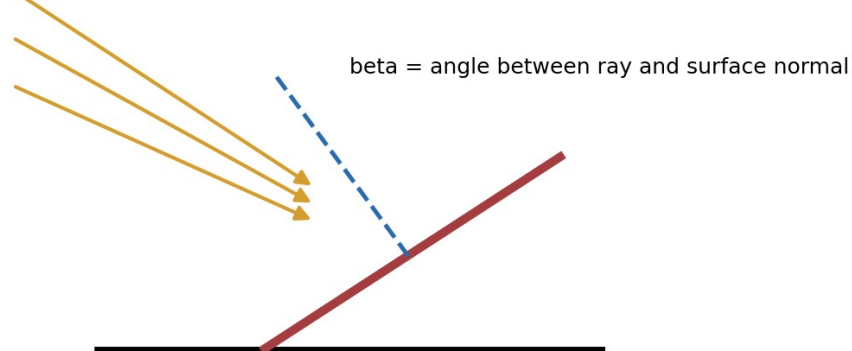
Priority: S - repeated | Asked: 2017, 2021, 2022, 2024 | Marks: 5-10

Year	Q.	Marks	Question focus
2017	7(a-i)	part	Short note: Cosine Law.
2021	6(b)	10	Atmospheric depletion and Cosine law.
2022	5(c)	5	Illustrate cosine law; use in building design.
2024	6(c)	7	Describe cosine law and applications in climate-responsive built environment.

$$I_{\text{surface}} = I_{\text{normal}} \times \cos \beta$$

β is measured between the incident solar ray and the NORMAL to the surface.

Cosine Law - oblique incidence spreads the same energy over a larger area



$$I_{\text{tilted}} = I_{\text{normal}} \times \cos(\beta)$$

Cosine law - oblique radiation spreads over a larger surface area.

Derivation idea

For a beam with cross-sectional area B normal to the rays, the same energy falls on a larger tilted surface area C . Geometry gives $B/C = \cos \beta$. Since intensity = energy/area, $I_C = I_B \times (B/C) = I_B \cos \beta$.

Building applications

- Compare solar exposure of roofs and façades at different orientations.
- Explain why low-angle incidence gives less energy per unit surface area but creates severe glare and deeper penetration through windows.
- Guide orientation and shading design.
- Estimate direct-beam component on a tilted surface.

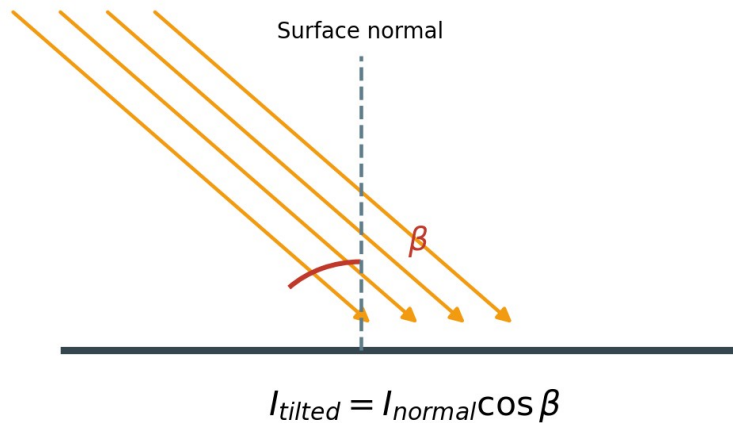
A3.4 Atmospheric Depletion and why higher altitude is colder

Priority: A/S - repeated | Asked: 2017, 2021, 2022, 2023 | Marks: 5-10

Year	Q.	Marks	Question focus
2017	7(a-ii)	part	Short note: Atmospheric Depletion.
2021	6(b)	10	Atmospheric depletion and Cosine law.
2022	6(b)	10	Atmospheric depletion; why colder at higher altitude?
2023	5(c)	5	Illustrate atmospheric depletion.

Atmospheric depletion is the reduction of solar radiation as it passes through the atmosphere by absorption and scattering by ozone, water vapour, dust, clouds and other particles. A longer atmospheric path generally causes greater depletion.

Cosine Law



Incidence and atmospheric path are separate effects on received solar energy.

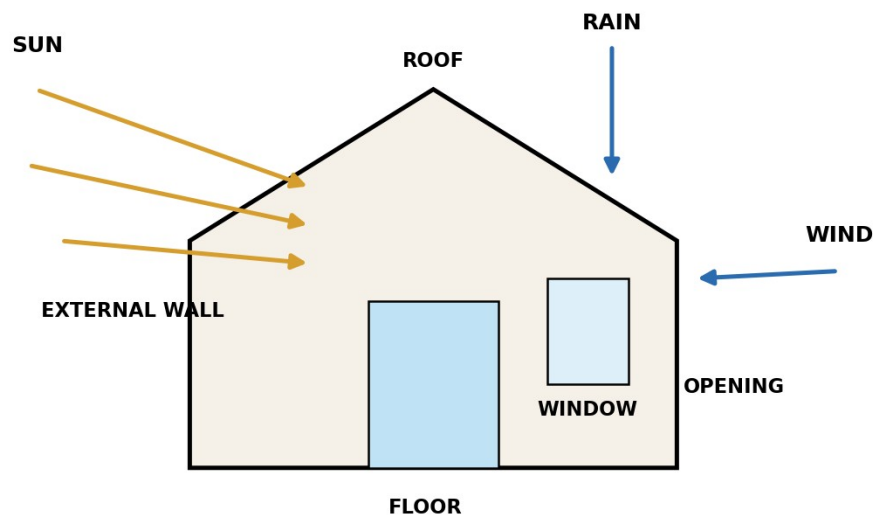
Higher altitude - exam-safe explanation

The lecture states that rising air becomes cooler. At lower pressure aloft, ascending air expands and cools, so average air temperature generally decreases with elevation. The atmosphere is also heated strongly from the ground upward, so being closer to the sun is not the controlling effect.

A3.5 Building envelope under climatic factors - roof, floor, wall, window, door

Priority: S - highest design repeat | Asked: 2018, 2021, 2022, 2023, 2024 | Marks: 10-22

Year	Q.	Marks	Question focus
2018	3(a)	22	Climate factors affecting building envelopes in designing microclimate.
2021	8(b)	18	Climatic factors affect building envelopes; sketches.
2022	8(b)	14	Building-envelope effects in microclimate.
2023	8(d)	12	Building envelope and climatic factors.
2024	8(b)	10	Building envelope and microclimate.



Microclimate design moderates heat, moisture, solar radiation and airflow at the envelope

Envelope as an environmental filter - heat, moisture, sun, rain and air.

Component	Main climatic problem	Design response
Roof	Solar heat gain, time lag, moisture penetration.	Reflective finish, insulation, air cavity/double roof, roof-level ventilation, waterproof membrane.
Ground floor	Ground heat exchange and damp.	DPC, moisture-resistant membrane, insulation where necessary.
External wall	Solar heat, rain, orientation-dependent exposure.	Material/thickness/colour/texture, insulation, overhangs, rain-shedding details.
Window	Direct solar gain, infiltration, rain/fog.	External shading, suitable glazing, controlled openings, weather-tight joints.
Ventilator / clerestory	Exhaust hot air, daylight and air change.	High-level openings and protected exhaust paths.
Door	Infiltration and access.	Weather-sealing where needed; use strategically for airflow.
Balcony / overhang	Sun + rain on wall/opening.	Projection shades and reduces rain exposure.
Terrace	Solar heating and re-radiation.	Cool/reflective finish, insulation, drainage, shade/planting.

Conclusion

Microclimate design is component-specific: each envelope part must filter the relevant combination of radiation, heat, moisture, rain and air movement rather than being treated as a purely structural surface.

A3.6 Design tools for moderating microclimate

Priority: A - repeated | Asked: 2018, 2022 | Marks: 5-10

Year	Q.	Marks	Question focus
2018	3(c)	5	Mention design tools for moderating microclimate.
2022	5(a)	part	Designer responsibilities + design tools for moderating microclimate.

Tool	What the designer manipulates
Materials	Type, thickness, composition, colour, texture, thermal and moisture properties.
Orientation	Relation of built form to sun and wind.
Openings	Size, position, fragmentation, louvers and shading.
Architectural design	Form, roof, section, courtyards, envelope.
Micro-planning	Place functions in suitable climatic zones.
Semi-mechanized techniques	Fans or controlled aids when passive response alone is insufficient.

A3.7 Microclimatic spacing solutions: warm-humid vs hot-dry

Priority: S - recent repeat | Asked: 2021, 2022, 2023 | Marks: 10

Year	Q.	Marks	Question focus
2021	7(b)	10	Microclimatic solution for hot-dry and warm-humid in terms of spacing.
2022	7(a)	10	Same family.
2023	7(b)	10	Same family.

Issue	Warm-humid	Hot-dry
Building spacing	Wide/open to admit breeze.	Closer/compact to create shade and reduce exposed area.
Arrangement	Staggered to reduce wind shadow.	Courtyard/cluster can create protected microclimate.
Outdoor surfaces	Vegetation and shaded porous surfaces.	Shade hard surfaces; evaporative cooling where water is viable.
Openings	Large and well distributed for cross flow.	Controlled; protect from hot dusty winds; night ventilation can be useful.

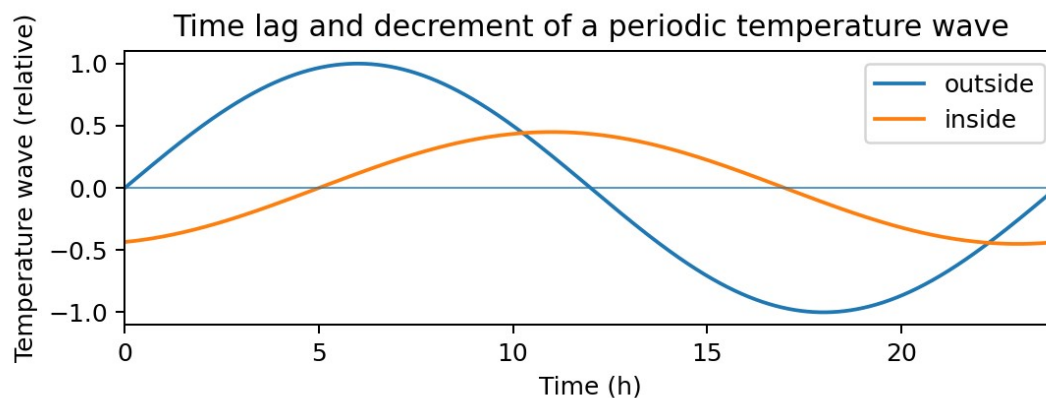
A3.8 Periodic heat flow, time lag, decrement factor, thermal capacity, U-value & solar gain factor

Priority: B - one paper but technical | Asked: 2018 | Marks: 10-15

Source status

The uploaded question bank names this topic, but the uploaded slides/notes do not contain the full derivation or numerical formula chain. The marked part below uses standard building-physics / solar-geometry knowledge as a supplementary method.

Year	Q.	Marks	Question focus
2018	8(b)	10	Periodic heat flow; time-lag and decrement factor.
2018	8(c)	15	Specific heat capacity vs thermal capacity; U-value; solar gain factor.



A heavy envelope delays and reduces the indoor temperature wave.

Term	Exam-ready definition
Periodic heat flow	Heat flow caused by periodically varying outdoor temperature/solar load, typically over a 24-hour cycle.
Time lag	Delay between the outdoor temperature/heat-flow peak and the corresponding indoor-surface peak.
Decrement factor	Ratio of indoor temperature-wave amplitude to outdoor/external wave amplitude; lower value means stronger damping.
Specific heat capacity c	Heat required to raise unit mass of material by 1 K; J/kg·K.
Thermal capacity	Total heat-storing capacity of an element; approximately $m \times c$ for a uniform body.
U-value	Overall steady heat-transfer coefficient through an assembly; lower U means better insulation, W/m ² ·K.
Solar gain factor	A factor relating incident solar radiation to heat admitted through a building element/opening; exact definition depends on the method used in class.

$$U = 1 / R_{\text{total}}$$

Supplementary standard relation; include internal/external surface resistances and layer resistances in R_{total} when your class method requires them.

Source status

The lecture/envelope material clearly discusses thermal capacity and time lag, but the uploaded slides do not give a complete formula set for decrement factor, U-value or solar-gain factor. The definitions and $U=1/R$ relation above are standard building-physics additions, explicitly marked supplementary.

A3.9 Lofty porch / extended overhead shade: summer protection, winter admission

Priority: A/S - recent | Asked: 2023, 2024 | Marks: 5-6

Year	Q.	Marks	Question focus
2023	6(a)	6	Lofty front porch / extended overhead shade: block summer glare and admit winter sun.
2024	5(c)	5	Same seasonal shading concept.

At a northern-hemisphere site, the summer sun is generally higher in the sky while winter sun is lower. A correctly sized horizontal overhang can intercept high-angle summer rays but allow low-angle winter rays to pass below the edge and enter the window.

Sketch instructions

9. Draw a vertical wall and window.
10. Draw an overhang above the window.
11. Draw a steep high summer ray just touching the outer overhang edge and falling outside the window.
12. Draw a shallow low winter ray passing under the overhang into the room.
13. Label: summer sun, winter sun, overhang, window, shaded zone.

Design caution

The exact projection depends on latitude, date/time and window geometry. The conceptual sketch alone is sufficient for the short 5-6 mark question; use A3.10 for numerical sizing.

A3.10 Solar shading numerical - 2023 and 2024 complete analytical solutions

Priority: A/S - numerical | Asked: 2023, 2024 | Marks: 15-17

Source status

The uploaded question bank names this topic, but the uploaded slides/notes do not contain the full derivation or numerical formula chain. The marked part below uses standard building-physics / solar-geometry knowledge as a supplementary method.

Year	Q.	Marks	Question focus
2023	7(a)	15	Khulna 22°N, 89°E; 3:30 pm 16 May 2024; SW window, normal 20° south of west; window 6 ft × 5 ft.
2024	7(b)	17	Khulna 22°N, 89°E; 2:00 pm 3 July 2025; SW window, normal 30° west of south; window 5 ft × 5 ft.

Method note

The uploaded slides contain cosine law and shading concepts but not the full numerical solar-position chain. The following is a standard analytical solar-geometry reconstruction. If your teacher uses a graphical sun-path chart, small numerical differences are expected; the method and shading logic remain the same.

Standard relations used

$$\text{HSA} = \text{Solar azimuth} - \text{Wall-normal azimuth}$$

$$\text{Profile angle } \Omega = \tan^{-1}(\tan \alpha / \cos \text{HSA})$$

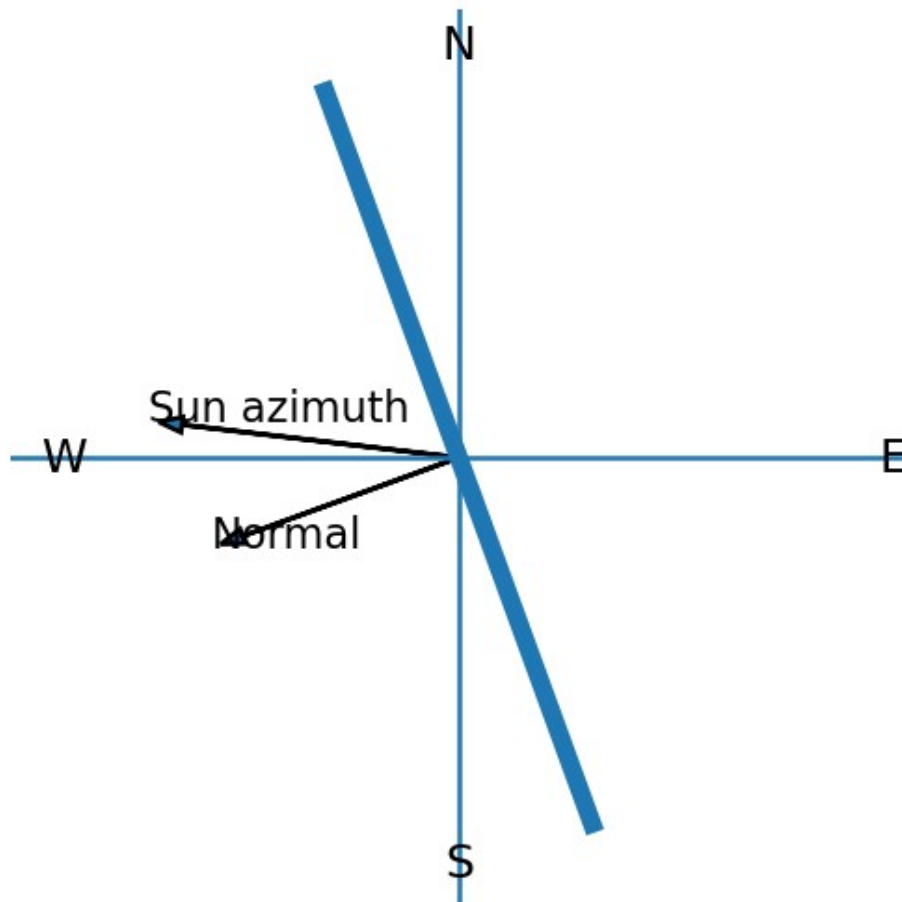
α = solar altitude.

$$\text{Horizontal overhang depth } P_h = H_{\text{window}} / \tan \Omega$$

$$\text{Vertical-fin depth } P_v = W_{\text{window}} / \tan |\text{HSA}|$$

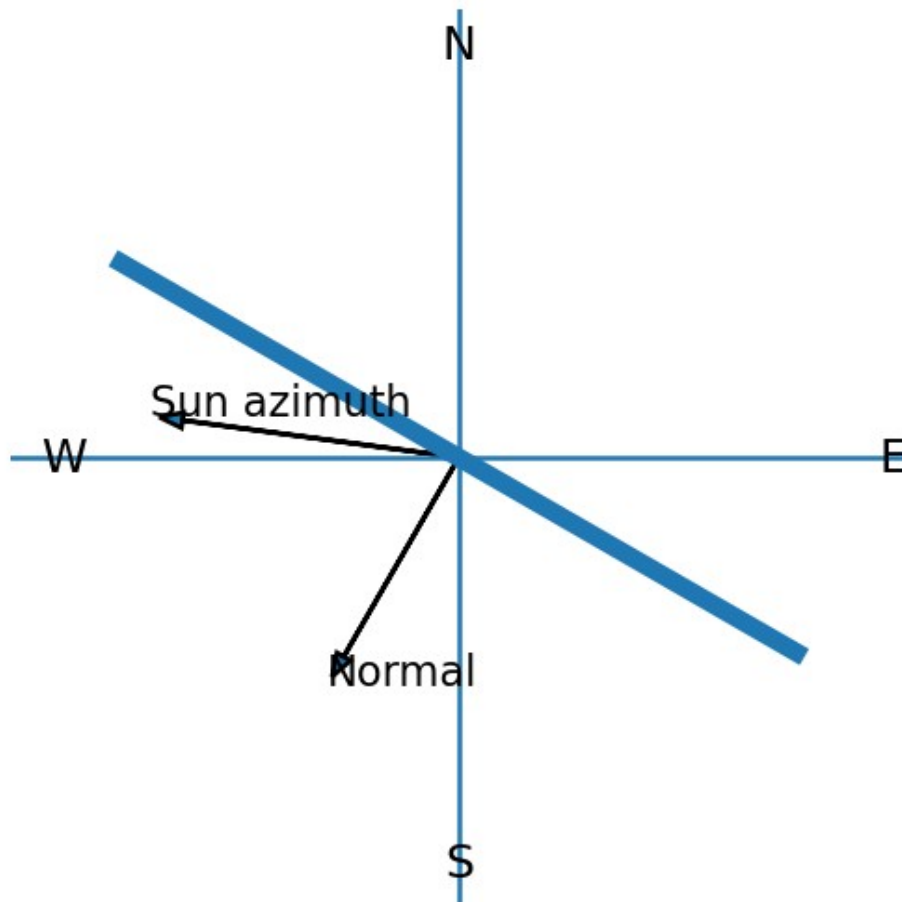
$$\cos \theta = \cos \alpha \times \cos \text{HSA}$$

θ = incidence angle from the wall normal for a vertical window.

2023 numerical result**2023_QB_Solar_Shading**

2023 orientation reconstruction: wall normal $\approx 20^\circ$ south of west (azimuth 250°).

Quantity	Result
Solar declination	19.18°
Equation of time	3.90 min
Local solar time	15.498 h
Solar altitude α	41.03°
Solar azimuth	276.78° from north
HSA	26.78°
Profile angle Ω	44.27°
Horizontal overhang depth	6.15 ft
Vertical fin depth	9.91 ft
Incidence angle θ	47.67°

2024 numerical result**2024_QB_Solar_Shading**

2024 orientation reconstruction: wall normal $\approx 30^\circ$ west of south (azimuth 210°).

Quantity	Result
Solar declination	23.03°
Equation of time	-3.87 min
Local solar time	13.869 h
Solar altitude α	64.12°
Solar azimuth	277.70° from north
HSA	67.70°
Profile angle Ω	79.57°
Horizontal overhang depth	0.92 ft
Vertical fin depth	2.05 ft
Incidence angle θ	80.47°

Orientation assumption

The 2023 plan is read as the window normal 20° south of west; the 2024 plan is read as 30° west of south. These readings come directly from the supplied plan sketches.

SET A4 - Climate, Context & Climate-friendly Construction

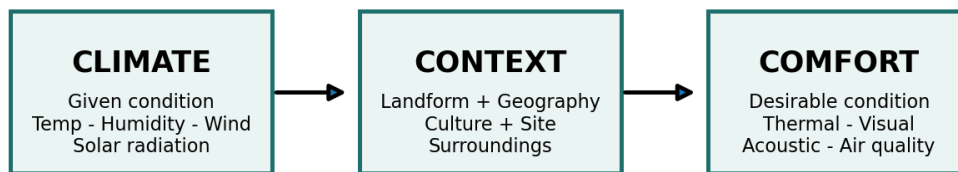
A4.1 Climate, Context, Comfort, Environment & Man-made Environment

Priority: S - broad repeat | Asked: 2016, 2017, 2018, 2024 | Marks: 5-15

Year	Q.	Marks	Question focus
2016	5(a)	15	Short notes: Context, Comfort, Man-made Environment.
2017	5(a)	10	Climate and Context from built-form designer viewpoint.
2017	6(a)	5	Difference between Climate and Environment.
2018	1(a)	10	Differences among Climate, Context and Comfort.
2024	5(a)	10	Differences among climate, context and comfort and design influence.

Term	Meaning	Design question
Environment	Overall physical surroundings and conditions in which people live.	What is around us?
Climate	Long-term environmental/weather condition - heat, humidity, wind, solar radiation, rain.	What environmental condition is given over time?
Context	Physical, geographic, social and cultural setting of the project.	Where and under what circumstances are we designing?
Comfort	Desired human condition - especially thermal, visual, acoustic and air-quality satisfaction.	What human state must the design achieve?
Man-made environment	Natural environment deliberately modified by people through buildings, infrastructure and landscape.	How have humans changed the given environment?

Climate-responsive design = respond to climate + context to achieve comfort



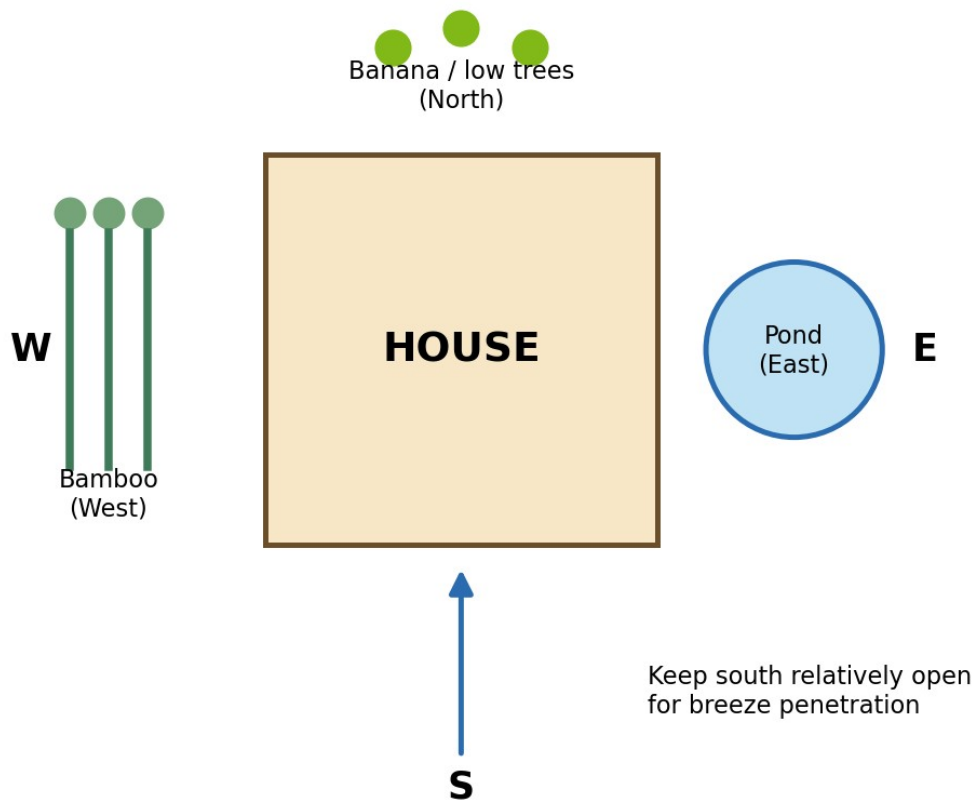
Climate = condition; Context = setting; Comfort = desired outcome.

A4.2 KHANA maxim and climate-responsive zoning

Priority: S - repeated | Asked: 2017, 2018, 2024 | Marks: 10-15

Year	Q.	Marks	Question focus
2017	5(b)	15	Khana: south best, east second, north burden, west curse - justify.
2018	2(a)	15	Zoning a building/site according to Khana in Bangladesh.
2024	5(b)	part	Relevance of Khana maxim in contemporary practice and functional zoning.

Vernacular climatic interpretation of KHANA maxim N



Simplified Khana-based climatic zoning for Bengal/Bangladesh context.

Traditional element / orientation	Climatic interpretation
Open / low obstruction to south	Allows useful southern breeze and solar access while avoiding major wind blockage.
Pond / water to east	Morning-side water and evaporative effect; excavated soil can raise plinth in flood-prone context.
Dense bamboo / vegetation to west	Screens low-angle afternoon west sun.
Low vegetation / banana to north	Maintains openness to southern/northern air path while using diffuse north light.

Exam caution

Present Khana as vernacular environmental wisdom in the climatic/cultural context of Bengal/Bangladesh, not as a universal modern code. Contemporary zoning should also consider actual site wind, solar path, drainage, privacy and regulations.

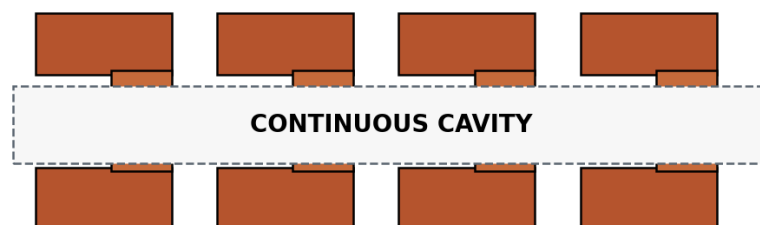
A4.3 Rat-trap bond: construction, working and climatic advantages

Priority: S - repeated | Asked: 2020, 2021, 2022, 2023 | Marks: 8-18

Year	Q.	Marks	Question focus
2020	2(a)	18	Construction technique of brick wall using Rat-Trap bond.
2021	7(d)	8	Working mechanism of rat-trap bond.
2022	7(b)	8	Working mechanism of rat-trap bond.
2023	8(c)	8	Working mechanism of rat-trap bond.

Rat-trap bond is a brick masonry technique in which bricks are placed on edge/vertical orientation so that a continuous cavity is formed within the wall thickness. The cavity reduces material use and improves thermal and acoustic resistance when the wall is properly detailed.

Rat-trap bond - bricks on edge create a cavity while retaining wall thickness



Exam sketch: show brick leaves, cavity, solid courses at sill/lintel/top/bottom

Rat-trap bond - cavity formed within a nominal brick-wall thickness.

Construction rules

- Use consistent good-quality bricks; dimensional variation affects the bond.
- Keep bottom/top closure courses and opening jamb/sill/lintel zones solid where required.
- Maintain the cavity without excessive mortar droppings bridging it.
- Provide reinforcement/bands/corner detailing where structural design or seismic requirements demand it.
- Plan services carefully so the cavity is not unintentionally destroyed.

Advantages

- Lower brick and mortar use; the lecture slide reports approximately 25% fewer bricks and 40% less mortar than conventional masonry in one comparison.
- Cavity improves thermal insulation: cooler inside in summer, warmer in winter.
- Lower material demand can reduce cost and embodied impact.
- Exposed brickwork can reduce plaster where workmanship is good.

Structural caution

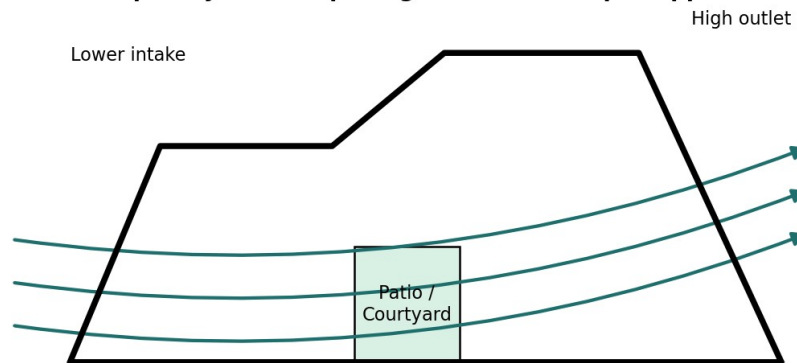
The slide says strength is not compromised. Exam-safe wording: properly designed and constructed rat-trap masonry can provide adequate wall performance; actual structural capacity depends on materials, workmanship, reinforcement/detailing and code requirements.

A4.4 Tube House: ventilation mechanism, spatial/climatic logic and modern reinterpretation

Priority: S - repeated | Asked: 2016, 2021, 2022, 2024 | Marks: 6-10

Year	Q.	Marks	Question focus
2016	8(b-i)	part	Tube House.
2021	7(c)	7	Tube-house air ventilation.
2022	7(c)	7	Tube-house air ventilation.
2024	8(c)	6	Tube house concept, spatial/climatic characteristics and modern reinterpretation.

Tube-house concept: adjustable openings + section shape support natural ventilation



Simplified tube-house ventilation concept.

A tube-house-type section uses a long/narrow spatial organisation, courts/voids and strategically placed low/high openings to move fresh air through occupied spaces and exhaust warmer air. The spatial sequence can support different day/night and private/semi-outdoor uses.

Modern urban reinterpretation

- Break long depth with courtyards/light wells so internal rooms are not sealed from daylight and air.
- Provide low inlet + high outlet to support cross/stack ventilation.
- Use shaded semi-outdoor transitional space rather than fully glazed heat-trapping corridors.
- Keep privacy through staggered openings/louvers while retaining airflow.
- Combine passive airflow with low-energy fans when urban obstruction weakens natural wind.

A4.5 Green Building, Sustainable Building & Organic Architecture

Priority: A/S - broad repeat | Asked: 2016, 2017, 2018, 2024/related | Marks: 5-16

Year	Q.	Marks	Question focus
2016	8(b-ii,iii)	part	Organic Architecture; Green building movement.
2017	8(a)	10	Define Green Building Movement.
2018	1(c)	10	Sustainable building vs green building.
2016	8(b)	part	Contemporary materials and climate-friendly movements.

Concept	Core meaning
Green building	Focuses on reducing measurable environmental/health impacts: energy, water, materials, waste and indoor environmental quality.
Sustainable building	Broader life-cycle balance of environmental, social and economic performance, resilience and long-term viability.
Organic architecture	Form and material respond to nature/site rather than treating nature as external decoration; integrate landscape, airflow, light and human use.
Climate-responsive design	Building form and envelope directly respond to local temperature, solar radiation, humidity, wind and rain.

Conclusion

A building can be “green” in selected performance areas yet still fall short of full sustainability if it is unaffordable, socially weak, short-lived or poorly adapted to future climate.

A4.6 Built-form designer responsibilities, complexities & "accept/reject/mitigate" position

Priority: A - repeated | Asked: 2017, 2018, 2020 | Marks: 10-20

Year	Q.	Marks	Question focus
2017	6(c)	20	Responsibilities and complexities in Climate Design.
2018	1(b)	15	Accept fate, reject development or mitigate?
2020	5(a)	15	Responsibilities of built-form designers and complexities.

Position

The climate-responsive position is mitigation: development should continue, but design should reduce environmental stress and energy dependence using orientation, form, envelope, shading, ventilation, landscape, materials and appropriate technology.

Complexity	Why it is difficult
Extreme climate	Comfort may not be achievable by openings alone.
Metabolic/internal heat	Occupancy and equipment add heat beyond outdoor climate.
Energy and cost	Comfort must be balanced with operational cost and maintenance.
Composite climate	Opposite seasonal strategies may be required.
Material availability/durability	Local skill, cost, moisture and weathering determine feasibility.

A4.7 Traditional vs contemporary building materials from climatic viewpoint

Priority: B - occasional | Asked: 2016, 2017, 2020 | Marks: 15-25

Source status

The uploaded question bank names this topic, but the uploaded slides/notes do not contain the full derivation or numerical formula chain. The marked part below uses standard building-physics / solar-geometry knowledge as a supplementary method.

Year	Q.	Marks	Question focus
2016	8(b-iv)	part	Contemporary building materials.
2017	8(b)	25	Compare contemporary and traditional building materials from climatic viewpoint.
2020	5(b)	15	Evaluate and compare contemporary and traditional building materials.

Aspect	Traditional/local materials	Contemporary/industrial materials
Thermal mass / insulation	Often climate-tuned through thick earth/brick, timber, thatch and shaded construction; performance varies widely.	Can be precisely engineered with insulation, membranes, low-e glazing, reflective coatings and composite systems.
Moisture response	Breathable materials may buffer moisture but can decay without protection.	Can provide strong waterproofing/vapour control, but wrong layer order can trap moisture.
Energy / transport	Often local and low processing, but durability/maintenance may be issues.	Higher embodied energy/transport can occur, but long life and high performance may offset operational energy.
Construction skill	Local craft knowledge is essential.	Requires technical detailing, quality control and compatible systems.
Climate suitability	Usually evolved for a known local climate.	Must be deliberately selected; "modern" does not automatically mean climate-responsive.

Source status

The uploaded climate-friendly slides name contemporary materials but do not provide a full comparison table. This balanced comparison is a supplementary synthesis using the climate-design principles taught in the course.

A4.8 Sustainability case: KUET academic building during heat wave

Priority: A - recent application | Asked: 2023, 2024 | Marks: 7-10

Year	Q.	Marks	Question focus
2023	8(a)	10	Heat-wave suspension; less light/air/social interaction; low cost/easy maintenance; analyze sustainability.
2024	8(d)	7	Same sustainability case.

Sustainability should be tested under three linked dimensions rather than only construction cost.

Dimension	Positive evidence	Negative evidence / improvement
Economic	Low construction cost; easy maintenance.	If heat waves stop academic operation, productivity and future retrofit/energy costs rise. Invest in passive upgrades with strong life-cycle value.
Environmental	Potentially low material/maintenance burden.	Poor daylight/ventilation can increase fan/AC demand and heat stress. Add shading, cross ventilation, roof/solar control, vegetation and daylight improvement.
Social	Provides academic accommodation at low cost.	Less light, poor air movement and lack of social-interaction space reduce health, learning, comfort and inclusiveness. Create shaded semi-outdoor/social spaces and better environmental quality.

Conclusion

The building is economically attractive in first cost but not fully sustainable if environmental failure disrupts use and social well-being. True sustainability requires balanced performance across all three dimensions.

A4.9 Vitruvius: climate and architecture suggestions

Priority: B - one direct paper | Asked: 2017 | Marks: 10

Year	Q.	Marks	Question focus
2017	5(c)	10	Suggestions of Vitruvius on climate and architecture.

The course presents Vitruvius as an early authority who argued that an architect must understand medicine/health in relation to region, atmosphere, location and water supply. His climate-related suggestions can be organized into eight recurring concerns:

No.	Concern	Design interpretation
1	Site	Select healthy site conditions.
2	Internal space	Plan spaces for environmental response.
3	Nature vs material	Select materials according to natural conditions.
4	Construction vs health	Construction should protect health.
5	Comfort vs temperament	Respond to human comfort and inhabitants.
6	Sustainability	Use resources rationally.
7	City planning	Arrange streets/buildings with sun and wind in mind.
8	Connectivity	Movement/environmental connection affects comfort.

SECTION B - COMPLETE SOLUTIONS | 4-set distribution

Preparation priority

Safest order: B1 → B3 → B2 → B4. B1 and B3 are exceptionally stable across the whole question bank.

SET B1 - Global Climatic Factors, Earth-Sun Relation & World Wind System

B1.1 Importance / professional use of climatic study for a BECM graduate

Priority: A/S - repeated | Asked: 2016, 2017 | Marks: 10-15

Year	Q.	Marks	Question focus
2016	1(a)	10	Importance of climate study for a BECM graduate.
2017	1(a)	15	Areas where a BECM professional can use learning from Climate & Architectural Design.

Use of climatic study	Professional design implication
Comfortable indoor climate	Set targets for temperature, humidity, airflow and solar control.
Energy-saving structural design	Use passive design first; reduce mechanical load.
Protection from external factors	Detail against sun, wind, rain, humidity and dust.
Use of natural resources	Exploit daylight, breeze, shade, vegetation and site features.
Regional architectural character	Adapt form, opening, roof and material to climate.
Site/material selection	Choose habitation zones and envelope systems based on local climatic risks.

B1.2 Major factors of Global Climate Variations

Priority: A/S - core list | Asked: 2016, 2024/fragmented | Marks: 25

Year	Q.	Marks	Question focus
2016	1(b)	25	Analyse five major factors of Global Climate Variations in different geographic locations.
2024	1-4	multiple	Tilt, ITCZ, wind pattern, radiation and site factors appear as split global-climate questions.

The lecture provides eleven global climatic factors. In a 25-mark “five factors” answer, select five and explain mechanism + geographic example.

No.	Global climatic factor
1	Solar radiation quality
2	Solar radiation quantity
3	Tilt of the Earth's axis
4	Radiation at the Earth's surface
5	Earth's thermal balance
6	Winds: thermal forces
7	Trade winds: Coriolis force
8	Mid-latitude westerlies
9	Polar wind
10	Annual wind shifts
11	Influence of topography

Five high-value mechanisms to explain

- Earth-Sun geometry/latitude changes solar angle and day length.
- Atmospheric depletion and angle of incidence modify radiation reaching the surface.
- Unequal heating creates pressure gradients and thermal circulation.
- Earth rotation produces Coriolis deflection and organizes trade/westerly/polar wind belts.
- Topography and land-water distribution modify regional temperature, humidity, rainfall and wind.

B1.3 Radiation at Earth Surface + Earth Thermal Balance + surface heat release

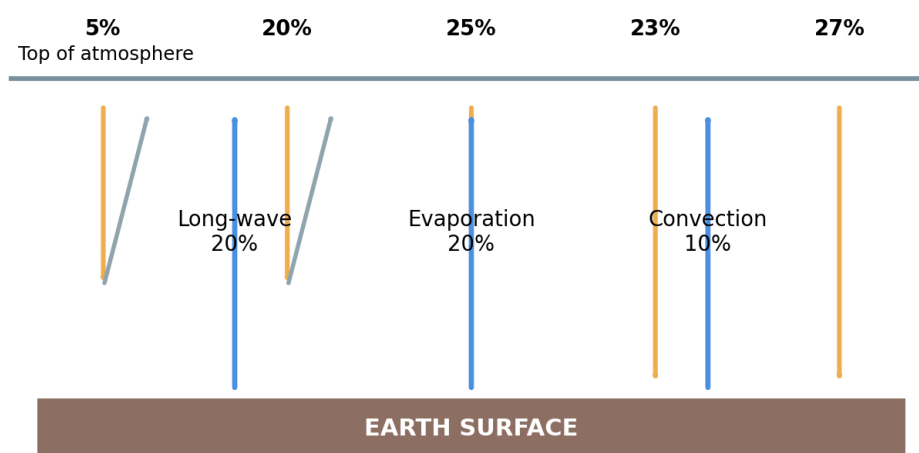
Priority: S - highest repeat | Asked: 2016, 2018, 2020, 2021, 2023, 2024 | Marks: 9-15

Year	Q.	Marks	Question focus
2016	2(b)	15	Earth thermal balance with diagram.
2018	5(a)	15	Ways Earth surface releases heat to atmosphere.
2020	6(a)	15	Earth thermal balance.
2021	4(c)	12	Earth thermal balance.
2023	2(a)	10	Radiation of Earth surface at global scale.
2024	4(c)	9	Radiation of Earth surface and global climate variation.

Radiation reaching the surface

- Angle of incidence / Cosine Law: oblique rays spread the same energy over a larger area.
- Atmospheric depletion: ozone, vapour, dust and clouds absorb/scatter radiation.
- Duration of sunshine: determines how long solar input can occur.

Incoming solar radiation: 50% reaches ground (23% diffuse + 27% direct)



Outgoing total from ground = 50%

Earth thermal balance - course percentage model.

Incoming component	Course value
Reflected from ground	5%
Reflected from cloud	20%
Absorbed in atmosphere	25%
Diffuse radiation reaching ground	23%
Direct radiation reaching ground	27%

Outgoing from ground	Course value
Long-wave radiation	20%
Evaporation, then radiation	20%
Convection, then radiation	10%
Total returned from ground	50%

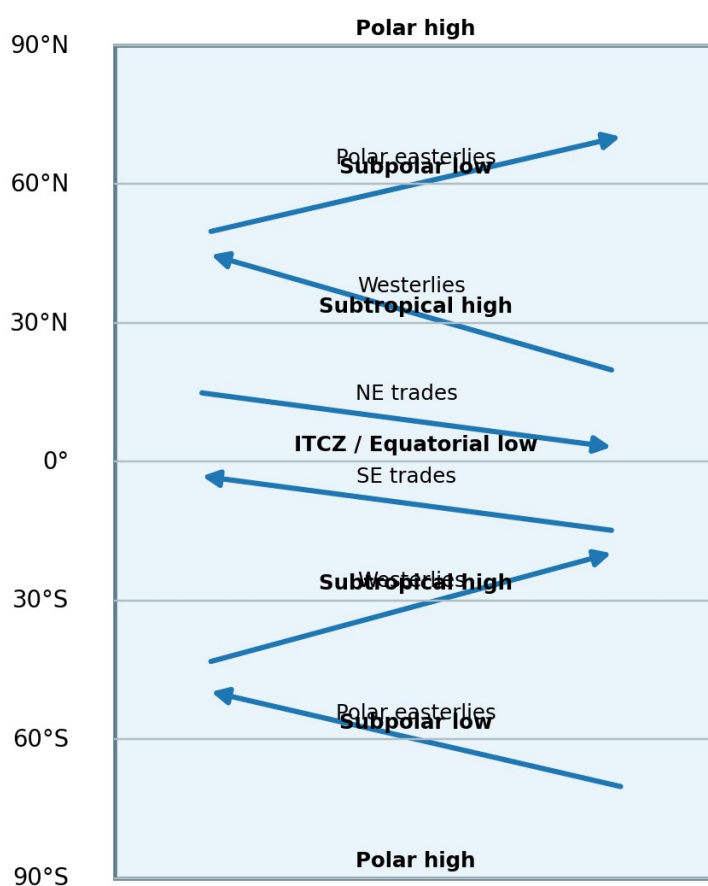
Thus, in the lecture model, 50% of incoming energy reaches the ground (23% diffuse + 27% direct) and the ground returns 50% through long-wave radiation, evaporation and convection. The surface therefore releases heat to the atmosphere mainly in three ways: radiation, latent heat through evaporation, and sensible heat through convection.

B1.4 Global wind pattern: thermal force, Coriolis, trade winds, westerlies & polar winds

Priority: S - very high | Asked: 2016, 2021, 2022, 2024 | Marks: 10-20

Year	Q.	Marks	Question focus
2016	3(a)	20	Forces interacting to generate global wind patterns at different latitudes.
2021	1(c)	15	Global wind patterns in both hemispheres.
2022	2(a,c)	10+13	Thermal/Coriolis force; trade winds, westerlies, polar winds.
2024	1(c)	15	Global wind patterns at different latitudes with sketches.

Global Wind Belts



Thermal force creates pressure belts; Coriolis force deflects the flow

Pressure belts, circulation cells and prevailing surface winds.

Unequal solar heating is the motive force: equatorial air becomes warm/light and rises, while colder dense air sinks in higher-pressure zones. Earth rotation then deflects moving air through the Coriolis effect.

Memory

0° low (ITCZ) ° 30high ° 60low ° 90high . Remember 0-30-60-90 first; the wind belts follow.

Latitude belt	Pressure / movement	Surface wind
0°	Strong heating; rising air; equatorial low.	Trade winds converge into ITCZ.
~30°	Air sinks; subtropical high.	Equatorward branch becomes trades; poleward branch becomes westerlies.
~60°	Westerly and polar flows converge; rising air; subpolar low.	Frontal/stormy zone.
90°	Cold dense sinking air; polar high.	Polar easterlies toward 60°.

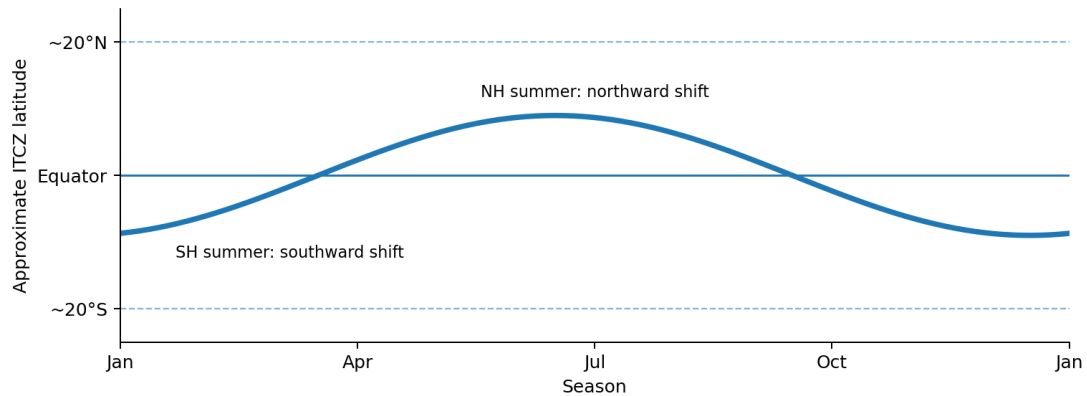
Coriolis directions

- Northern Hemisphere: moving air deflects to the right.
- Southern Hemisphere: moving air deflects to the left.
- Therefore trades are easterly, mid-latitude winds are westerly and polar surface winds are easterly.

B1.5 ITCZ / Doldrums and Annual Wind Shift

Priority: S - very high | Asked: 2016, 2021, 2023, 2024 | Marks: 10-15

Year	Q.	Marks	Question focus
2016	3(b)	15	Impact of annual wind shift from north to south of equator.
2021	1(b)	10	Relationship between ITCZ and annual wind shifts.
2023	2(b)	10	Doldrums: location and formation.
2024	1(b)	12	Why ITCZ is calm and why it shifts more toward Northern Hemisphere.



Seasonal migration of the ITCZ - conceptual annual shift around the equator.

The Inter-Tropical Convergence Zone (ITCZ) is the equatorial/sub-equatorial belt where the NE and SE trade winds converge. Warm moist air rises, so surface horizontal wind can be weak, variable and calm - the traditional "doldrums".

Why it shifts

- The zone of maximum solar heating migrates north and south with the apparent seasonal movement of the sun.
- ITCZ follows this thermal low, moving northward in northern summer and southward in northern winter.
- It often migrates farther north over the Northern Hemisphere because there is more land, and land heats more strongly/quickly than ocean in summer, pulling the thermal low northward.

Effects of the annual shift

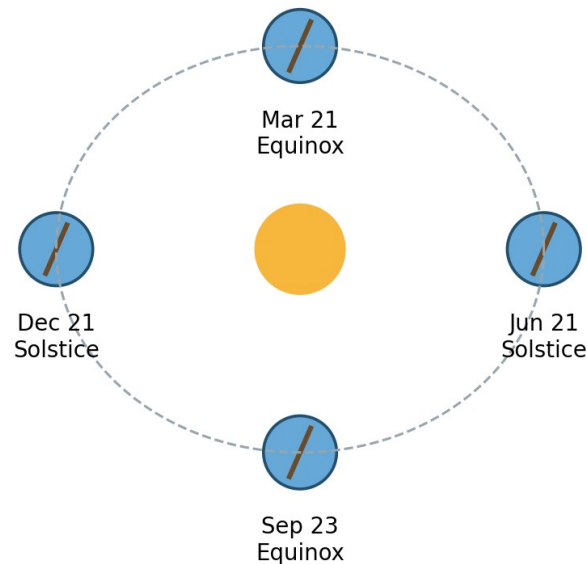
- Seasonal reversal/shift of prevailing winds.
- Monsoon circulation and seasonal rainfall migration.
- Changes in cloud, humidity and storm activity.
- Important seasonal design changes in ventilation and rain protection.

B1.6 Earth-Sun Relationship, tilt of axis and seasonal solar distribution

Priority: S - high | Asked: 2017, 2022, 2024 | Marks: 8-20

Year	Q.	Marks	Question focus
2017	1(b)	20	Tilt of Earth axis and global climate variation.
2022	2(b)	12	Earth-Sun Relationship and global climate variation.
2024	1(a)	8	How tilt affects solar radiation distribution and seasons.

Earth-Sun Relationship: tilt controls seasonal solar distribution



Earth-Sun geometry: solstices, equinoxes and axial tilt.

- Earth axis is about 23.5° from the perpendicular to the orbital plane (equivalent to 66.5° to the orbital plane in the lecture wording).
- Because the tilted Earth revolves around the Sun, solar angle and day length change by latitude and season.
- Around 21 June the overhead-sun zone reaches the Tropic of Cancer; around 21 December it reaches the Tropic of Capricorn.
- Around 21 March and 23 September, solar declination is near the equator (equinox).
- Higher sun angle concentrates radiation; lower angle spreads energy over larger area and increases atmospheric path.

B1.7 Influence of Topography on climate variation

Priority: A - repeated | Asked: 2021, 2023, 2024/related | Marks: 5-10

Year	Q.	Marks	Question focus
2021	1(a)	10	Influence of topography on global climate variations.
2023	4(a)	5	Topographical factors responsible for global climate variation.
2024	3(a)	related	Local site deviations also depend on topography.

- Altitude: temperature generally decreases with height; mountain climates differ from valleys.
- Slope/aspect: changes solar exposure and therefore surface heating.
- Orographic effect: mountains force moist air upward, supporting cloud/rain on windward sides and drier conditions in rain-shadow zones.
- Cold-air drainage: dense cool air can settle in depressions/valleys and produce local inversions.
- Wind modification: hills, ridges and valleys block, accelerate, channel or create turbulence in airflow.

Conclusion

Topography does not create the global circulation by itself; it modifies the general climate into distinct regional and site climates.

SET B2 - Tropical Climate Classification & Climate-responsive Design

B2.1 Warm-humid vs Hot-dry climate - comparison

Priority: S - highest repeat | Asked: 2016, 2018, 2022, 2024 | Marks: 10-20

Year	Q.	Marks	Question focus
2016	2(a)	20	Compare Hot-dry with Warm-humid.
2018	5(b,c)	10+10	Major tropical characteristics; importance to designers.
2022	1(b)	15	Composite/monsoon and hot-dry desert patterns.
2024	2(b,c)	16+12	Warm-humid equatorial; hot desert formation/day-night range.

Feature	Warm-humid equatorial	Hot-dry desert / semi-desert
Location	Roughly within 15° N/S.	Approximately 15-30° N/S.
Temperature range	High but small diurnal/annual range.	Very hot days, much cooler nights; large diurnal range.
Humidity	High, often ~75%.	Low, roughly 10-55% in lecture profile.
Rainfall	Heavy, frequent; cloudy/wet.	Very low and irregular; long dry periods.
Sky / solar	Cloudy; diffuse radiation important.	Mostly clear; strong direct sun and glare/dust haze.
Wind	Generally low except squalls.	Hot, local, dusty winds; dust storms possible.
Vegetation	Rapid/lush growth, high moisture.	Sparse vegetation, dry soil.
Primary design challenge	Remove heat + moisture; maximize shade and airflow.	Reduce daytime heat gain, use thermal mass/time lag, shade and controlled ventilation.

B2.2 Why passive control is critical in tropical climates

Priority: S - repeated | Asked: 2017, 2022, 2023, 2018/related | Marks: 5-15

Year	Q.	Marks	Question focus
2017	2(a)	15	Passive control in tropical climate more critical - justify.
2022	1(a)	5	Same statement.
2023	1(d)	5	Same statement.
2018	5(c)	10	Tropical climates of maximum importance for built-form designers - reasons.

- Tropical climates often impose a persistent heat problem, so an unresponsive building accumulates heat for long periods.
- Solar control, ventilation and moisture management can be achieved structurally before mechanical systems are sized.
- Passive measures operate whenever climate occurs, reducing energy and peak cooling load.
- Warm-humid and hot-dry climates need opposite strategies; therefore the built form itself must be climate-specific.
- Power reliability, cost and maintenance make passive resilience important in many tropical regions.

Conclusion

The statement is reasonable: passive design is not “optional decoration” in the tropics; it is the first layer of thermal protection and ventilation, with active systems used as support.

B2.3 Composite / Monsoon climate - characteristics and examples

Priority: S - repeated | Asked: 2017, 2021, 2022 | Marks: 10-20

Year	Q.	Marks	Question focus
2017	2(b)	20	Characterize composite or monsoon climate with example.
2021	2(c)	10	Composite/monsoon climate pattern.
2022	1(b)	15	Composite/monsoon and hot-dry desert patterns.

Composite/monsoon climate occurs over large tropical/subtropical land masses where solar and wind regimes change strongly by season. The lecture describes a combination of hot-dry and warm-humid periods, sometimes with a cool-dry season.

Season	Typical lecture profile	Design implication
Hot-dry	High day temperature, lower RH, dusty/hot winds.	Shade, thermal mass/time lag, controlled hot wind.
Warm-humid / monsoon	High RH, strong prolonged rain, cloudy sky, fairly strong monsoon wind.	Cross ventilation, rain protection, moisture control.
Cool-dry (where present)	Cooler nights and lower humidity.	Admit useful sun, reduce excessive ventilation/drafts.

Examples shown in the lecture include Lahore, Mandalay, Asuncion, Kano and New Delhi.

B2.4 Classification of tropical climate and climatic-zone formation

Priority: S - repeated | Asked: 2021, 2022, 2023 | Marks: 10-15

Year	Q.	Marks	Question focus
2021	2(a)	10	Atmospheric factors affecting Atkinson classification.
2022	1(c)	15	How climatic zones are formed on earth + ecological consideration.
2023	1(a)	10	Classification of tropical climate and formation of climatic zones.
2023	3(a)	10	How climatic zones are formed with examples.

The lecture credits G. A. Atkinson (1953) and bases tropical classification mainly on temperature and humidity extremes that affect human comfort.

Six named tropical climate types
1. Warm-humid equatorial climate
2. Warm-humid island / trade-wind climate
3. Hot-dry desert / semi-desert climate
4. Hot-dry maritime desert climate
5. Composite / monsoon climate
6. Tropical upland climate

How climatic zones form

- Latitude controls average solar angle and solar receipt.
- Earth-Sun seasonal geometry changes day length and solar intensity.
- Global pressure/wind belts distribute heat and moisture.
- Land-water contrast changes seasonal heating and humidity.
- Altitude/topography modify temperature, rainfall and wind, creating regional variations.

B2.5 Warm-humid equatorial climate

Priority: A/S - recent | Asked: 2023, 2024 | Marks: 10-16

Year	Q.	Marks	Question focus
2023	1(b)	10	Features of warm-humid equatorial climate with examples.
2024	2(b)	16	Why climate is warm and humid between equator and 15° N/S.

Parameter	Lecture profile
Location	Roughly within about 15° N/S.
Air temperature	Mean max roughly 27-32°C; mean min 21-27°C; small diurnal/annual range.
RH	High, around 75% and may vary roughly 55-100%.
Rainfall	Very high, roughly 2000-5000 mm/year in lecture profile; intense storms possible.
Sky	Cloudy, often 60-90% cloud cover; glare possible.
Solar	Cloud-reflected/scattered; diffuse component important.
Wind	Generally low, though squalls can be strong.
Vegetation / risks	Rapid growth, high water table/waterlogging; mould, algae, rust, rot, insects.

It remains warm because of high solar receipt near the equator, and humid because abundant evaporation, moisture convergence and frequent rainfall keep atmospheric water-vapour content high.

B2.6 Climate-specific protection: Warm-humid measures + Hot-dry desert formation, character & precautions

Priority: S - high | Asked: 2021, 2022, 2023, 2024 | Marks: 10-15

Year	Q.	Marks	Question focus
2021	2(b)	10	Protective measures in warm-humid climate regions.
2022	1(b)	15	Hot-dry desert pattern.
2023	1(c)	10	Precautions while constructing in hot-dry desert climate.
2024	2(c)	12	How desert forms in subtropical region and why day-night range is large.

2021: protective measures in warm-humid climate

- Provide maximum ventilation and air movement with large, well-positioned openings.
- Provide maximum shading against both direct and diffuse solar radiation.
- Avoid unnecessary heat storage; favour light construction where appropriate.
- Use reflective/light-coloured outer surfaces and well-insulated, ventilated double roofs.
- Use vegetation for shade and microclimate moderation without blocking useful breezes.
- Protect walls/openings from heavy rain and moisture while keeping airflow paths open.

Hot-dry desert: why it forms and why day-night range is large

- Around 20-30° latitude, descending air in the subtropical high suppresses cloud formation and rainfall.
- Clear skies allow intense daytime solar heating and strong night-time long-wave radiation.
- Low humidity and sparse vegetation mean little latent-heat buffering; surfaces heat and cool rapidly.

Hot-dry building precautions

- Use compact form and mutual shading to reduce exposed surface area.
- Use thick/heavy envelope where useful to exploit thermal mass and time lag.
- Keep openings small/controlled during hot dusty periods; use protected night ventilation when outdoor air becomes cooler.
- Shade roofs, walls and openings; use light/reflective external surfaces.
- Use courtyard and evaporative cooling where water availability permits.
- Protect against dust and sand infiltration with screens, vestibules and careful detailing.

Source status

Warm-humid measures are taken directly from the uploaded design-specification material. Hot-dry protective measures synthesize the tropical-climate characteristics with the uploaded Section A envelope/design principles; this cross-source synthesis is marked rather than presented as a separate Billah-slide list.

B2.7 Ecological considerations for built-environment design

Priority: B/A - asked twice | Asked: 2016, 2022 | Marks: 10-15

Source status

The uploaded question bank names this topic, but the uploaded slides/notes do not contain the full derivation or numerical formula chain. The marked part below uses standard building-physics / solar-geometry knowledge as a supplementary method.

Year	Q.	Marks	Question focus
2016	4(b)	10	Major issues of Ecological Consideration for a built-environment designer.
2022	1(c)	part of 15	Ecological consideration taken care of by built-environment designers.

The uploaded introductory slide names “Ecological consideration” but does not provide a detailed list. The following exam-safe expansion is therefore supplementary and stays within climate-responsive design principles:

- Minimize unnecessary disturbance of landform, drainage and natural water movement.
- Preserve useful vegetation and use landscape for shade/wind without blocking beneficial ventilation.
- Reduce heat-island effects through shaded/reflective/permeable surfaces.
- Conserve water and manage rainwater/stormwater.
- Use local/durable/low-impact materials where feasible.
- Reduce operational energy through passive climate response before mechanical systems.
- Protect indoor health while avoiding pollution transfer to the external environment.
- Design for resilience to flood, heat, wind, rainfall and future climatic stress.

B2.8 Atkinson classification factors / basis

Priority: B - one direct paper | Asked: 2021 | Marks: 10

Year	Q.	Marks	Question focus
2021	2(a)	10	Atmospheric factors affecting classification of tropical climates as suggested by G.A. Atkinson.

Atkinson's tropical classification in the lecture is based on extremes/levels of temperature and humidity as they affect human comfort, producing warm-humid, hot-dry and seasonal/upland groupings. In an exam, state the climatic variables explicitly and then list the six named types from B2.4.

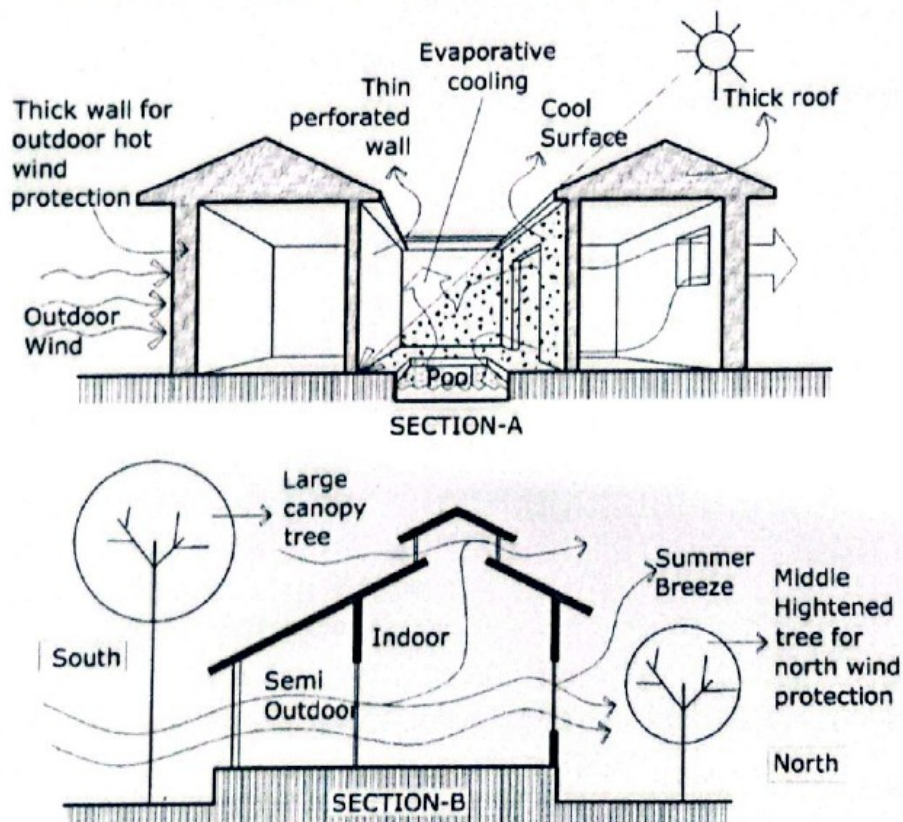
B2.9 Traditional built forms: identify climates and explain design response

Priority: B/A - recent application | Asked: 2024 | Marks: 12

Year	Q.	Marks	Question focus
2024	8(a)	12	Two traditional built forms: identify two climates and analyze components, surroundings and orientation.

The following two sections represents two distinct traditional built forms that are (1: climatically suitable for two distinct regional climatic conditions.

- Identify the two types of climates represented by each built form by analyzing the sections.
- Analyze how the building components, site surroundings and orientations respond effectively to their respective climatic conditions.



Given built-form sketches from the 2024 question paper.

Section-A sketch: Hot-dry climate

- Thick wall / thick roof provide heat capacity and delay daytime heat penetration.
- Perforated wall allows controlled airflow while screening hot wind/glare.
- Pool and wetted/cool surfaces provide evaporative cooling.
- Protected courtyard-like arrangement limits direct solar exposure.

Section-B sketch: Warm-humid climate

- Large canopy tree provides shade without enclosing the building.
- Open/semi-outdoor spaces admit summer breeze.
- High roof / high-level opening helps warm air escape.
- Vegetation can filter undesirable north wind while allowing useful cross ventilation.
- Light/open form reduces heat storage and supports continuous air movement.

Conclusion

The two forms illustrate opposite priorities: hot-dry design stores/delays heat and creates shade/evaporative cooling, while warm-humid design minimizes heat storage and maximizes shaded ventilation.

SET B3 - Regional, Site & Urban Climate

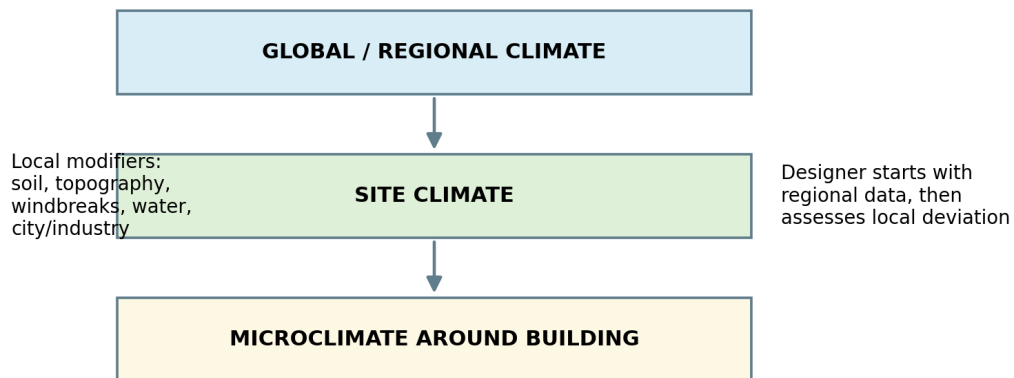
B3.1 Site climate / microclimate: factors and local deviations

Priority: S - highest repeat in B3 | Asked: 2016, 2017, 2018, 2020, 2021, 2022, 2023, 2024 | Marks: 10-25

Year	Q.	Marks	Question focus
2016	4(a)	25	Local factors developing site/microclimate vs regional/macro.
2017	4(a,b)	10+15	Microclimate factors; site-climate factors.
2018	6(a)	20	Site climate and factors.
2020	3(b),6(b)	20/15	Factors forming site climate / local factors.
2021	4(b)	10	Site climate and responsible factors.
2022	4(a)	10	Site vs regional climate.
2023	3(b)	10	Site-climate design process.
2024	3(a)	10	Site climate and factors causing local deviation.

Site climate is the climate of the actual project area, considered in both horizontal extent and height. It is usually a local modification of the broader regional/macrocimate, and regional meteorological data is only a starting point.

Climate scale and deviation



Regional climate → local modifiers → site / microclimate.

Hierarchy	Factors
Primary	Global climatic factors; the slide highlights temperature, humidity and airflow as primary climatic outcomes.
Secondary	Location of region; diurnal variation; altitude.
Tertiary	Soil condition; topography; natural/man-made windbreakers; water bodies; cities/industries; deserts.
Local & minor	Ocean current; sky condition; smog; smoke; fog; thunder/lightning; sand storms.

Direct site-climate factors

- Soil condition in locality
- Topography of the site
- Small hills/rocks as windbreakers
- Man-made windbreakers
- Small water bodies
- Cities/industries and built mass

B3.2 Site climate vs Regional / Macroclimate

Priority: S - repeated | Asked: 2017, 2021, 2022 | Marks: 5-15

Year	Q.	Marks	Question focus
2017	3(a)	15	Differentiate Site Climate from Regional Climate.
2021	3(a)	10	Urban climate vs macroclimate.
2022	4(a)	10	Site climate vs regional climate.

Regional / Macroclimate	Site / Microclimate
Covers a broad geographic region; commonly represented by meteorological-station data.	Covers the actual project site or very local area.
Dominated by global and regional climatic processes.	Modified strongly by topography, surface, vegetation, water, buildings and windbreaks.
Useful general guide but may miss local deviations.	Required for final orientation, zoning, opening, landscape and envelope decisions.
Changes slowly over distance at regional scale.	Can change over very short distance and height.

B3.3 Local modifiers: ocean current, windbreakers, soil, topography, smog/fog

Priority: A - short-note support | Asked: 2017, 2020, 2024/related | Marks: 10-20

Year	Q.	Marks	Question focus
2017	4(c)	10	Short notes: Ocean current; Natural wind breaker.
2017	7(b)	10	Tertiary factors constituting regional climate.
2020	3(b)	20	Five factors contributing to site climate.

Factor	Effect
Soil condition	Stony/sandy surfaces heat and re-radiate rapidly; fine/organic soil behaves differently.
Topography	Cold heavy air may settle in depressions; slopes alter solar/wind exposure.
Natural windbreaker	Mountains/trees obstruct, redirect or filter wind.
Ocean current	Transports heat and modifies coastal temperature, humidity and rainfall.
Water body	Moderates temperature and changes local humidity/breeze.
Smog/smoke/fog	Modify solar radiation, visibility, humidity/air quality.
Sand storm	Requires protected openings and dust-resistant detailing.

B3.4 Temperature inversion

Priority: B/A - asked 2018/2021 | Asked: 2018, 2021 | Marks: 10-15

Source status

The uploaded question bank names this topic, but the uploaded slides/notes do not contain the full derivation or numerical formula chain. The marked part below uses standard building-physics / solar-geometry knowledge as a supplementary method.

Year	Q.	Marks	Question focus
2018	6(b)	15	Temperature inversion with example.
2021	3(b)	10	Temperature inversion from perspective of global warming.

The uploaded site-climate slide lists temperature inversion as a deviation factor but does not fully define its mechanism. Standard meteorological meaning: a temperature inversion occurs when temperature increases with height over a layer instead of normally decreasing. Cool dense air can be trapped near the ground beneath warmer air.

Typical causes / example

- Night-time radiative cooling of ground under clear calm conditions can cool the near-surface air.
- Cold air can drain into valleys/depressions and become trapped.
- A warm layer above suppresses vertical mixing, allowing fog/smog and pollutants to accumulate.

Exam caution

Do not claim “global warming causes all inversions”. Treat inversion as a vertical/local atmospheric structure that can intensify urban pollution/thermal discomfort; the question wording “from the perspective of global warming” can be answered by discussing its interaction with urban heat/pollution rather than a direct one-to-one cause.

B3.5 Fundamental process of designing a site / residential building using site-climate knowledge

Priority: S - repeated | Asked: 2020, 2022, 2023 | Marks: 10

Year	Q.	Marks	Question focus
2020	3(a)	10	Designing a large site using site-climate knowledge.
2022	4(b)	10	Fundamental process of designing a site.
2023	3(b)	10	Residential building in KUET using site-climate knowledge.

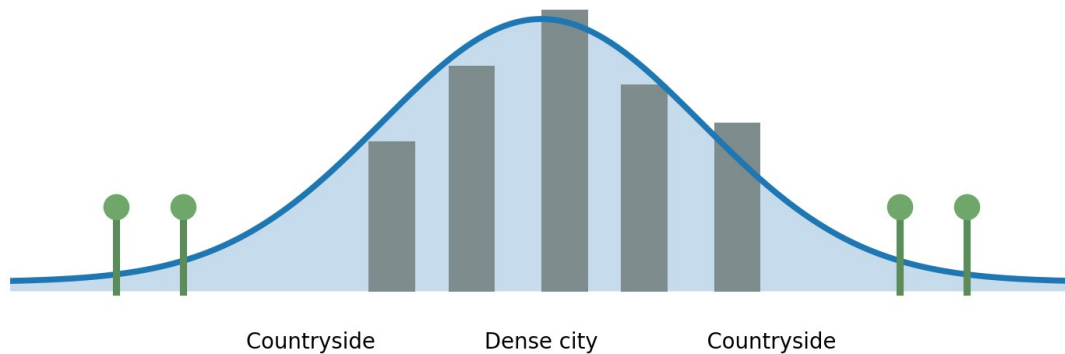
14. Collect regional climatic data as the baseline.
15. Identify dominant local modifiers: topography, water, vegetation, existing buildings, soil and windbreaks.
16. Estimate or measure site deviations in temperature, wind, humidity, sun/rain exposure.
17. Observe seasonal extremes and, where long records are unavailable, consult experienced local observers.
18. Select the most suitable habitation/building zone within the site.
19. Orient and zone the building to useful wind/solar access and risks.
20. Select envelope, openings, shading and landscape based on the site findings.
21. Check whether the design moderates adverse conditions without destroying beneficial local microclimate.

B3.6 Urban climate: human intervention, deviation and effects

Priority: S - high | Asked: 2021, 2022, 2023, 2024 | Marks: 8-15

Year	Q.	Marks	Question focus
2021	3(a)	10	Urban climate vs macroclimate.
2022	4(c)	15	Factors and effects of urban-climate deviation.
2023	3(c)	15	Urban climate deviation factors and effects.
2024	3(b)	8	Urban climate depends on human intervention - explain.

Urban climate: heat storage + less evaporation + anthropogenic heat



Conceptual urban heat-island profile.

Human intervention	Urban-climate effect
Changed surface qualities - pavement/buildings	Higher solar absorption and less evaporation.
Building mass and geometry	Cast shadows, block/channel wind, store heat and release it slowly.
Energy seepage from AC, appliances, engines, industry	Adds anthropogenic heat.
Atmospheric pollution	Reduces outgoing radiation and air quality; can intensify haze/smog.

Effects reported in lecture

- City-countryside temperature deviation up to about 8°C.
- Urban RH about 5-10% lower in the stated slide example due to quick runoff and less vegetation.
- Wind velocity may fall to less than half of nearby countryside.
- Strong turbulence/eddies can form near leeward building corners.

SET B4 - Elements of Climate, Measurement, DRI & Basic Thermal Response

B4.1 Driving Rain and Driving Rain Index (DRI)

Priority: S - highest repeat | Asked: 2021, 2022, 2023, 2024 | Marks: 5-10

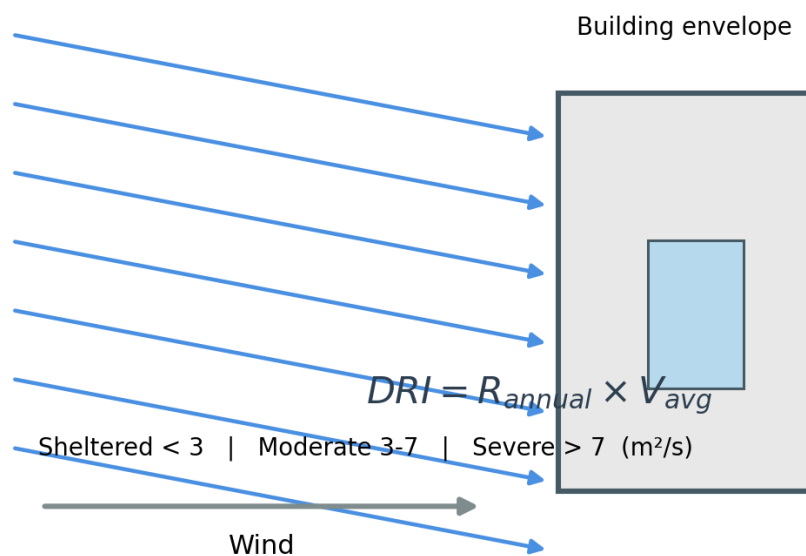
Year	Q.	Marks	Question focus
2021	3(c-iii)	5	Short note: Driving rain.
2022	3(c)	10	Driving rain + DRI building exposure.
2023	2(c)	10	Driving rain + DRI.
2024	2(a)	7	Driving rain + building exposure by DRI.

Driving rain is wind-driven rain that strikes a façade at an angle rather than falling vertically. The oblique impact increases the chance of water penetration at joints, cracks, window reveals and other envelope gaps.

$$\text{DRI} = \text{Annual Rainfall (m)} \times \text{Annual Average Wind Speed (m/s)}$$

Lecture unit: m^2/s .

Driving rain = wind-driven rain striking envelope at an angle



Driving rain mechanism and exposure classification.

Exposure	DRI	Design response
Sheltered	< 3 m^2/s	Normal weather detailing; adequate sills, flashings and joints.
Moderate	3-7 m^2/s	More careful reveals, joint sealing, overhangs and rain screen/drainage.
Severe	> 7 m^2/s	Robust envelope, protected openings, drained cavities/rain screen and strong gap control.

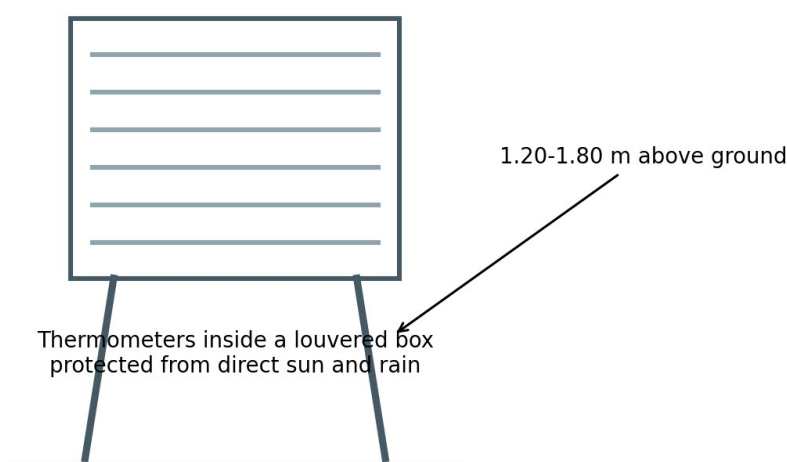
Slide example: 1.2 m annual rainfall $\times$ 4.0 m/s average wind = 4.8 m^2/s $\rightarrow$ Moderate exposure.

B4.2 Measuring climate: temperature, humidity, wind + Stevenson Screen

Priority: S - repeated | Asked: 2017, 2018, 2021, 2022, 2024 | Marks: 8-20

Year	Q.	Marks	Question focus
2017	7(a-iii)	part	Short note: Stevenson Screen.
2018	3(b)	8	Working mechanism of Stevenson Screen.
2021	3(c-ii)	5	Short note: Stevenson Screen.
2022	3(b)	10	Instrument for temperature data in open field; working mechanism.
2024	7(c)	10	Measure temperature, humidity and airflow at a specific site.

STEVENSON SCREEN



Stevenson Screen - shaded, ventilated enclosure for true air-temperature instruments.

Parameter	Instrument / method	Important field condition
Air temperature	Dry-bulb mercury/alcohol/electronic thermometer; continuous record by thermograph.	Measure in shade inside louvered white Stevenson screen, around 1.2-1.8 m above ground.
Humidity	Wet-and-dry bulb hygrometer or electronic hygrometer; psychrometric interpretation.	Record with matching temperature; analyse mean/extreme RH.
Wind speed	Cup/propeller anemometer; continuous anemograph.	Free wind normally ~10 m over open terrain; urban readings may be 10-20 m to reduce obstruction bias.
Wind direction	Wind vane / anemograph.	Record direction frequency and speed class; summarize in wind rose.

Stevenson Screen working logic

- White finish reflects solar heat.
- Louvers permit free air circulation but shade instruments from direct radiation.
- Double roof reduces direct heating from sun and protects from rain.
- Raised stand reduces immediate ground heat influence.

B4.3 Human thermal comfort - Section B fundamentals: objective/subjective variables and balance

Priority: S - frequent overlap | Asked: 2018, 2021, 2022, 2023, 2024 | Marks: 5-20

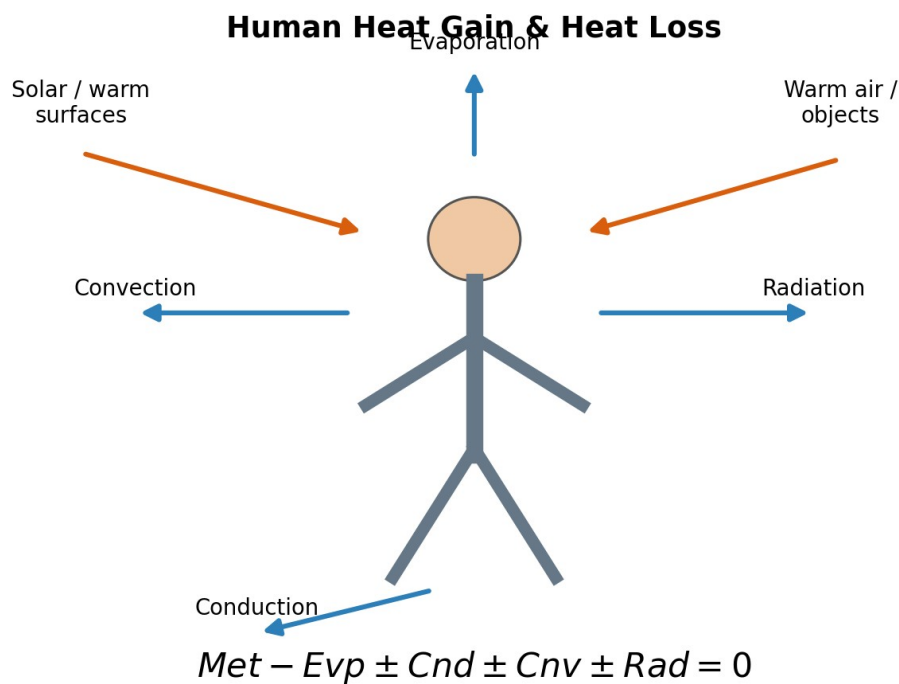
Year	Q.	Marks	Question focus
2018	7(a,b,c)	20+5+10	Thermal comfort; subjective variables; effective temperature.
2021	4(a)	13	Thermal balance equation.
2022	3(d)	10	Heat production + subjective variables.
2023	4(b,c,d)	8+10+12	Objective variables; significance; thermal balance.
2024	3(c,d)	12+5	Thermal comfort/body balance; subjective variables.

Section B thermal-comfort answer uses the same physiological foundation as A1 but emphasizes classification of variables and the natural/individual distinction.

Objective / climatic variables	Subjective / individual variables
Radiation	Clothing
Air temperature	Acclimatization
Humidity	Age and sex
Air movement	Body shape / subcutaneous fat
	State of health, food/drink, skin colour

$$M - W = C + R + K + E + S$$

At steady state $S \approx 0$.



Body heat balance and physiological response.

Memory

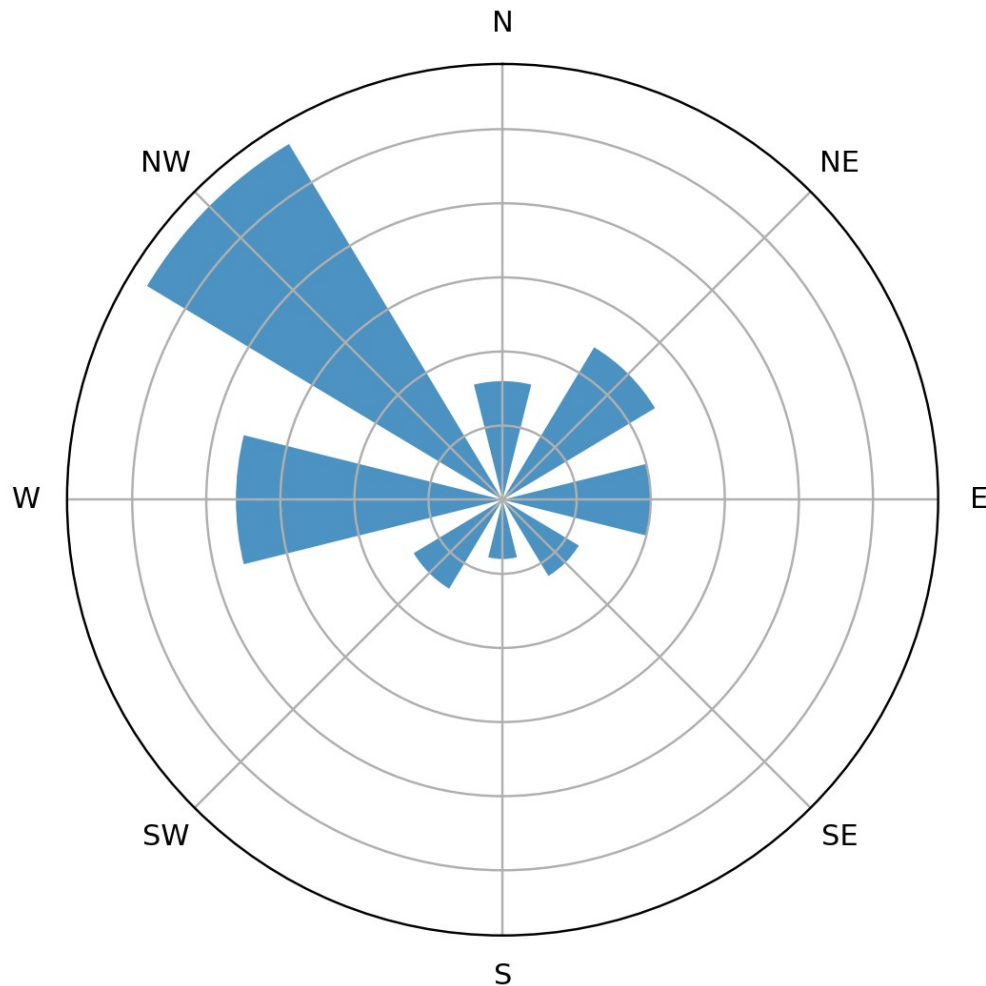
Objective = what the environment does; Subjective = what the person brings.

B4.4 Monthly and Annual Wind Frequency Graph / Wind Rose

Priority: A/S - repeated | Asked: 2017, 2024 | Marks: 13-20

Year	Q.	Marks	Question focus
2017	3(b)	20	Draw monthly and annual wind frequency graphs and interpret.
2024	4(a)	13	Draw monthly and annual wind frequency graph and explain interpretation.

Wind Rose: direction + frequency



Wind rose: direction, frequency, speed classes and calm percentage.

- Each spoke points toward the direction FROM which wind arrives.
- Spoke length represents frequency of occurrence from that direction.
- Segment/line thickness or colour can represent speed classes.
- Calm percentage may be shown at the centre.
- A monthly graph shows seasonal change; the annual graph aggregates the whole year and identifies the dominant prevailing directions.

Design interpretation

- Orient openings and building spacing to desirable seasonal winds.
- Protect façades from harmful hot/dusty/cold winds.
- Locate service/buffer zones on adverse wind sides.
- Combine wind-frequency data with driving-rain exposure before locating openings.

B4.5 Elements of climatic information and how climatic factors affect building design

Priority: A/S - repeated | Asked: 2021, 2024 | Marks: 5-13

Year	Q.	Marks	Question focus
2021	2(d)	5	Outline elements of climatic information.
2024	4(b)	13	Climatic factors affecting building design and how they influence it.

Element	Design influence
Temperature	Heat gain/loss, thermal mass, insulation and comfort.
Humidity / vapour pressure	Evaporation, condensation, mould/decay and comfort.
Precipitation / driving rain	Roof pitch, drainage, joints, waterproofing and protected openings.
Solar radiation	Orientation, shading, glazing, roof/wall heat gain and daylight.
Wind	Ventilation, infiltration, pressure, structural exposure and rain penetration.
Sky condition	Diffuse radiation/daylight and solar availability.
Vegetation	Shade, evapotranspiration, wind filtering and surface moisture.

Designer workflow

Record → Analyse means/extremes → Identify beneficial/harmful features → Translate into form, orientation, envelope, openings, shading and landscape.

B4.6 Humidity factors and reciprocal climate relationships

Priority: B/A - one direct paper | Asked: 2020 | Marks: 15

Year	Q.	Marks	Question focus
2020	1 (b)	15	Humidity depends on soil, waterbody, altitude, precipitation, air movement and vegetation; show reciprocal relationship.

Factor	Effect on humidity / related climate
Soil condition	Wet/vegetated soil supports evaporation; dry/stony surfaces tend to contribute less local moisture.
Water body	Evaporation increases atmospheric moisture and moderates temperature.
Altitude	Temperature falls with height; cooling can raise RH and support condensation/cloud.
Precipitation	Adds surface moisture and wetness; subsequent evaporation can raise humidity.
Air movement	Transports moist/dry air and controls evaporation rate.
Vegetation	Evapotranspiration adds moisture; shade lowers surface temperature and modifies local RH.

The relationships are reciprocal: humidity changes evaporation, cloud and precipitation; those processes in turn modify radiation, temperature and airflow, so the climatic elements interact rather than operate independently.

B4.7 Atmospheric Pressure

Priority: B - short repeated | Asked: 2022, 2023 | Marks: 5

Year	Q.	Marks	Question focus
2022	3(a)	5	Briefly describe atmospheric pressure.
2023	2(d)	5	Briefly describe atmospheric pressure.

Atmospheric pressure is the force per unit area exerted by the weight of the air column above a surface. In moist air it can be considered as the sum of the partial pressure of dry air and the partial pressure of water vapour.

$$P = P_a + P_v$$

P_a = partial pressure of dry air; P_v = vapour pressure.

- Pressure generally decreases with altitude.
- Pressure differences drive wind from higher toward lower pressure, then Coriolis and friction modify direction.
- Heating can make air less dense and contribute to rising motion/low-pressure development.

B4.8 Solar radiation quantity / duration vs intensity

Priority: B/A - short support | Asked: 2021, 2024/related | Marks: 5

Year	Q.	Marks	Question focus
2021	3(c-i)	5	Short note: Solar radiation quantity.
2024	4(c)	related	Radiation at Earth surface.

The course slide gives a solar constant of about 1395 W/m^2 at the upper atmosphere and notes variations from solar output and Earth-Sun distance. At a building site, distinguish sunshine duration from radiation intensity: duration tells how long bright sun occurs; intensity is actual radiant power per unit area.

Quantity	Typical unit / instrument
Sunshine duration	hours/day; Campbell-Stokes sunshine recorder.
Radiation intensity	W/m^2 ; pyranometer/pyrheliometer/actinometer/solarimeter.

NUMERICAL QUICK SHEET - final revision

Topic	Formula / result
Relative Humidity	$RH = AH/SH \times 100$; $10/17 \times 100 = 58.8\%$.
Ventilation ACH	$ACH = Q \times 3600 / V$.
Air-change heat loss	$Q_v = nV/3600$; Heat = $\rho Q_v c_p \Delta T$.
2021/2024 heat-loss case	≈ 3.015 kW with $\rho = 1.20$ kg/m ³ .
2022 heat-loss case	≈ 1.884 kW with $\rho = 1.20$ kg/m ³ .
2023 heat-loss case	≈ 2.513 kW with $\rho = 1.20$ kg/m ³ .
2021 room ventilation (200 persons)	1200 L/s = 1.20 m ³ /s; ACH ≈ 1.39 .
2022 room ventilation (150 persons)	900 L/s = 0.90 m ³ /s; ACH ≈ 1.04 .
2023 room ventilation (180 persons)	1080 L/s = 1.08 m ³ /s; ACH ≈ 1.15 using $45 \times 25 \times 3$.
DRI	Annual rainfall (m) $\times$ annual average wind speed (m/s).

PAPER-BY-PAPER COVERAGE CHECKLIST

Purpose

This appendix is the “no-skipping” check. Each original paper sub-question is mapped to one or more master solution IDs above. Historical Section A/B labels are not used for sorting here; the mapping follows the current 8-set distribution.

2016

Q.	Focus	Solution ID
1(a)	Importance of climate study	B1.1
1(b)	Five major global climate variation factors	B1.2
2(a)	Hot-dry vs warm-humid	B2.1
2(b)	Earth thermal balance	B1.3
3(a)	Forces generating global wind	B1.4
3(b)	Annual wind shift	B1.5
4(a)	Local/site microclimate factors	B3.1
4(b)	Ecological consideration	B2.7
5(a)	Context, comfort, man-made environment	A4.1
5(b)	Human body heat exchange	A1.1
6(a)	Factors of climatic comfort	A1.2
6(b)	Thermal comfort ideas	A1.2
6(c)	ET, CET, bioclimatic chart	A1.4
7(a)	Heat exchange process for built environment	A3.5
7(b)	Thermal balance equation	A1.3
7(c)	Effective shading	A3.2
8(a)	Types of shading devices	A3.2
8(b-i)	Tube House	A4.4
8(b-ii)	Organic Architecture	A4.5
8(b-iii)	Green Building Movement	A4.5
8(b-iv)	Contemporary building materials	A4.7

2017

Q.	Focus	Solution ID
1(a)	Professional use of climate course	B1.1
1(b)	Tilt of Earth axis	B1.6
2(a)	Passive control in tropical climate	B2.2
2(b)	Composite/monsoon climate	B2.3
3(a)	Site vs regional climate	B3.2
3(b)	Monthly/annual wind-frequency graph	B4.4
4(a)	Microclimate and factors	B3.1
4(b)	Site climate and factors	B3.1
4(c-i)	Ocean current	B3.3
4(c-ii)	Natural windbreaker	B3.3
5(a)	Climate & Context	A4.1
5(b)	Khana maxim	A4.2
5(c)	Vitruvius	A4.9
6(a)	Climate vs Environment	A4.1
6(b)	Essential knowledge for climate design	A1.8
6(c)	Designer responsibilities/complexities	A4.6
7(a-i)	Cosine Law	A3.3
7(a-ii)	Atmospheric Depletion	A3.4
7(a-iii)	Stevenson Screen	B4.2
7(a-iv)	Supersaturated air	A2.5
7(a-v)	Corrected Effective Temperature	A1.4
7(b)	Tertiary factors of regional climate	B3.3
8(a)	Green Building Movement	A4.5
8(b)	Contemporary vs traditional materials	A4.7

2018

Q.	Focus	Solution ID
1(a)	Climate, Context, Comfort	A4.1
1(b)	Accept/reject/mitigate	A4.6
1(c)	Sustainable vs green building	A4.5
2(a)	Khana zoning	A4.2
2(b)	Layout/spacing/openings for ventilation	A3.1
3(a)	Building envelope & microclimate	A3.5

Q.	Focus	Solution ID
3(b)	Stevenson Screen	B4.2
3(c)	Design tools for moderating microclimate	A3.6
4(a)	Heat loss in climatic situations	A1.5
4(b)	Human body heat exchange	A1.1
5(a)	Ways Earth surface releases heat	B1.3
5(b)	Major characteristics tropical climate	B2.1/B2.4
5(c)	Importance of tropical climates to designers	B2.2
6(a)	Site climate factors	B3.1
6(b)	Temperature inversion	B3.4
7(a)	Thermal comfort/heat production-release	B4.3
7(b)	Subjective variables	B4.3
7(c)	Effective temperature limitations	A1.4
8(a)	Passive vs active thermal control	A1.7
8(b)	Periodic heat flow/time lag/decrement	A3.8
8(c)	Specific heat capacity/Thermal capacity/U-value/Solar gain factor	A3.8

2020

Q.	Focus	Solution ID
1(a)	Human body heat exchange	A1.1
1(b)	Humidity factors and interaction	B4.6
2(a)	Rat-trap bond	A4.3
2(b)	AH/SH/RH numerical	A2.5
3(a)	Large-site design using site climate	B3.5
3(b)	Five site climate factors	B3.1
4(a)	Active vs passive thermal control	A1.7
4(b)	Heat loss climatic situations	A1.5
5(a)	Designer responsibilities/complexities	A4.6
5(b)	Contemporary vs traditional materials	A4.7
6(a)	Earth thermal balance	B1.3
6(b)	Local/site microclimate factors	B3.1

2021

Q.	Focus	Solution ID
1(a)	Topography influence	B1.7
1(b)	ITCZ and annual wind shift	B1.5
1(c)	Global wind pattern	B1.4
2(a)	Atkinson tropical classification factors	B2.8
2(b)	Warm-humid protective measures	B2.6
2(c)	Composite/monsoon climate	B2.3
2(d)	Elements of climatic information	B4.5
3(a)	Urban vs macroclimate	B3.2/B3.6
3(b)	Temperature inversion	B3.4
3(c-i)	Solar radiation quantity	B4.8
3(c-ii)	Stevenson Screen	B4.2
3(c-iii)	Driving rain	B4.1
4(a)	Human thermal balance	B4.3
4(b)	Site climate factors	B3.1
4(c)	Earth thermal balance	B1.3
5(a)	Air change/ventilation expression	A2.1
5(b)	Heat loss climatic situations	A1.5
5(c)	Designer responsibilities thermal comfort	A1.6
5(d)	Room ventilation numerical	A2.2
6(a)	Air-space/infiltration + heat-loss numerical	A2.3
6(b)	Atmospheric depletion + Cosine law	A3.3/A3.4
6(c)	ET/CET/EW	A1.4
6(d)	AH/SH/RH	A2.5
7(a)	Fire shelter numerical	A2.4
7(b)	Hot-dry vs warm-humid spacing	A3.7
7(c)	Tube-house ventilation	A4.4
7(d)	Rat-trap bond	A4.3
8(a)	Indoor climatic comfort by shading	A3.2
8(b)	Building envelope/microclimate	A3.5
8(c)	Laboratory ventilation numerical	A2.6

2022

Q.	Focus	Solution ID
1(a)	Passive control tropical	B2.2
1(b)	Composite/monsoon + hot-dry desert	B2.3/B2.6
1(c)	Climatic-zone formation + ecological consideration	B2.4/B2.7
2(a)	Thermal force + Coriolis	B1.4
2(b)	Earth-Sun relationship	B1.6
2(c)	Trade wind + westerlies + polar winds	B1.4
3(a)	Atmospheric pressure	B4.7
3(b)	Temperature measurement / Stevenson Screen	B4.2
3(c)	Driving rain / DRI	B4.1
3(d)	Human heat production + subjective variables	B4.3
4(a)	Site vs regional climate	B3.2
4(b)	Large-site design process	B3.5
4(c)	Urban-climate deviation/effects	B3.6
5(a)	Designer responsibilities + microclimate tools	A1.6/A3.6
5(b)	Heat loss climatic situations	A1.5
5(c)	Cosine law application	A3.3
5(d)	Room ventilation numerical	A2.2
6(a)	Air-space/infiltration + heat-loss numerical	A2.3
6(b)	Atmospheric depletion + altitude	A3.4
6(c)	ET/CET/EW/OT	A1.4
6(d)	Vasomotor adjustment	A1.9
7(a)	Microclimatic spacing hot-dry/warm-humid	A3.7
7(b)	Rat-trap bond	A4.3
7(c)	Tube house	A4.4
7(d)	Fire shelter numerical	A2.4
8(a)	Comfort by shading	A3.2
8(b)	Building envelope	A3.5
8(c)	Laboratory ventilation	A2.6
8(d)	Air-space/Air-change/Infiltration short notes	A2.1

2023

Q.	Focus	Solution ID
1(a)	Tropical classification + climate zones	B2.4
1(b)	Warm-humid equatorial	B2.5
1(c)	Hot-dry desert precautions	B2.6
1(d)	Passive tropical control	B2.2
2(a)	Radiation at Earth surface	B1.3
2(b)	Doldrums	B1.5
2(c)	Driving rain / DRI	B4.1
2(d)	Atmospheric pressure	B4.7
3(a)	Climatic-zone formation	B2.4
3(b)	KUET residential/site-climate design	B3.5
3(c)	Urban-climate deviation/effects	B3.6
4(a)	Topographical factors	B1.7
4(b)	Objective thermal-comfort variables	B4.3
4(c)	Significance of thermal comfort	B4.3
4(d)	Thermal-balance regulation	B4.3
5(a)	Vasomotor adjustment	A1.9
5(b)	ET/CET/EW/OT	A1.4
5(c)	Atmospheric depletion	A3.4
5(d)	Air-space/infiltration heat loss	A2.3
6(a)	Lofty porch/seasonal shading	A3.9
6(b)	Natural/cross ventilation and block spacing	A3.1
6(c)	Heat loss/gain climatic conditions	A1.5
6(d)	Room ventilation numerical	A2.2
7(a)	Solar shading numerical	A3.10
7(b)	Microclimatic spacing	A3.7
7(c)	Fire shelter numerical	A2.4
8(a)	Sustainability case	A4.8
8(b)	Laboratory ventilation	A2.6
8(c)	Rat-trap bond	A4.3
8(d)	Building envelope	A3.5

2024

Q.	Focus	Solution ID
1(a)	Earth-axis tilt and seasons	B1.6
1(b)	ITCZ calmness and northward shift	B1.5
1(c)	Global wind pattern	B1.4
2(a)	Driving rain / DRI	B4.1
2(b)	Warm-humid equatorial climate	B2.5
2(c)	Subtropical desert + diurnal range	B2.6
3(a)	Site climate/local deviations	B3.1
3(b)	Urban climate/human intervention	B3.6
3(c)	Thermal comfort/body balance	B4.3
3(d)	Subjective variables	B4.3
4(a)	Wind-frequency graph	B4.4
4(b)	Climatic factors affecting design	B4.5
4(c)	Radiation at Earth surface	B1.3
5(a)	Climate/Context/Comfort	A4.1
5(b)	Khana relevance and zoning	A4.2
5(c)	Lofty porch seasonal shading	A3.9
5(d)	Laboratory ventilation	A2.6
6(a)	Vasomotor adjustment + design	A1.9
6(b)	Designer thermal-comfort responsibilities + essential knowledge	A1.6/A1.8
6(c)	Cosine law applications	A3.3
6(d)	Fire shelter numerical	A2.4
7(a)	Air-space/infiltration + heat loss	A2.3
7(b)	Solar shading numerical	A3.10
7(c)	Measuring temperature/humidity/airflow	B4.2
8(a)	Traditional built forms / two climates	B2.9
8(b)	Building envelope	A3.5
8(c)	Tube house modern reinterpretation	A4.4
8(d)	Sustainability case	A4.8

Source Base & Final Exam Notes

- Regular Final Term Question Bank: 2016, 2017, 2018, 2020, 2021, 2022, 2023 and 2024 scans.
- Section A: uploaded class notes, lecture slides 1-11 and student lecture-wise question analysis.
- Section B: uploaded Billah Sir / Abu Safayet Sir lecture slides and class notes.
- Reference books: Manual of Tropical Housing and Building; A Text Book of Climate & Design for the Built-form Designers.
- Where the source itself is ambiguous, the ambiguity is written in the solution instead of being silently changed.

Last-minute rule

For each 10-20 mark theory answer: Definition → mechanism/factors → table → sketch → design significance → conclusion.
For each numerical: units first, formula second, substitution third, safety/design interpretation last.

Made by 2223033